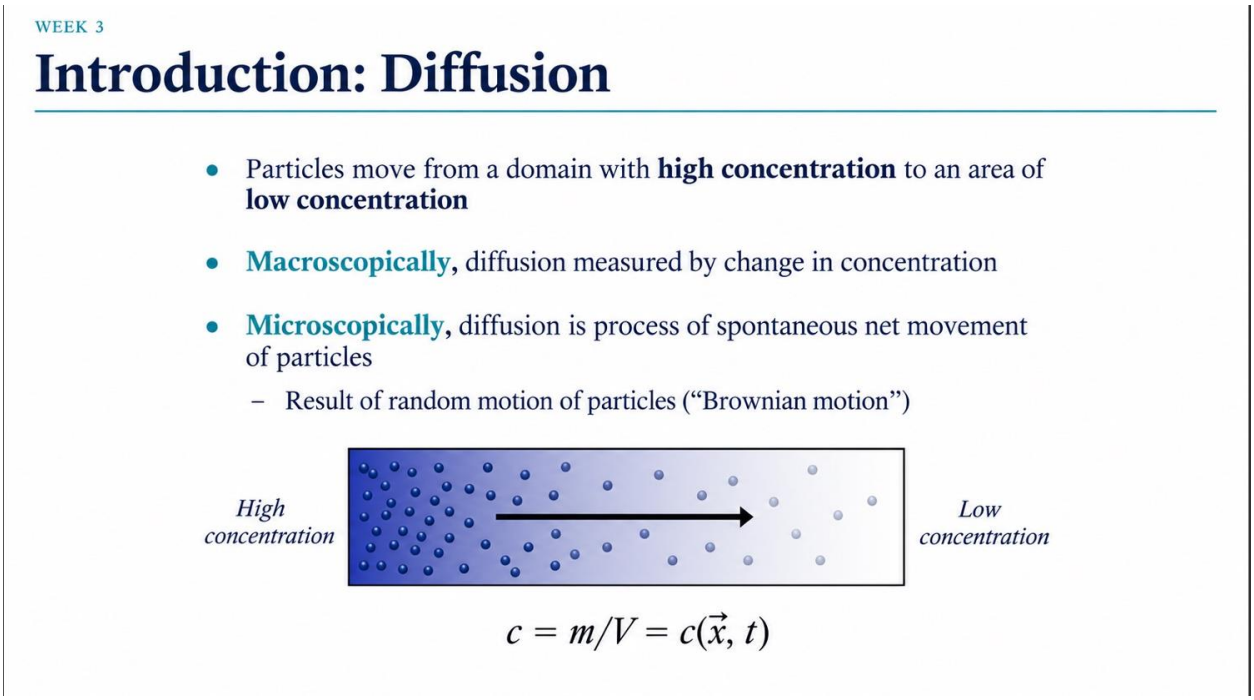
Week 3

Computer Modeling



Slide 1: Introduction: Diffusion

- **Definition:** Particles naturally move from a region of **high concentration** to an area of **low concentration**.
- **Two Perspectives:**
    - **Macroscopic:** We measure diffusion by observing overall changes in concentration across space and time.
    - **Microscopic:** It is a net particle movement resulting from **Brownian motion**—the individual, random movement of particles.
- **Formula:** $c = m/V = c(\vec{x}, t)$

- c: Concentration, calculated as mass (m) divided by volume (V).

- $c(\vec{x}, t)$: Expresses that concentration changes based on location ($\vec{x}$) and time (t).

WEEK 3

Ink droplet in water



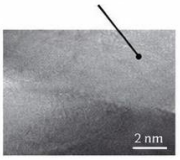
WEEK 3

Relevant scales in materials

λ(atom) << ξ(crystal) << d(grain) << dx, dy, dz << H, W, D



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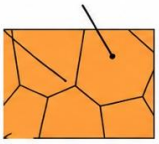


Fig. 8.7 in: Buehler, M. *Atomistic Modeling of Materials Failures*. Springer, 2008.

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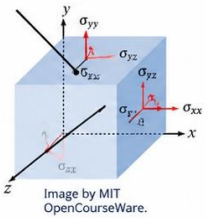


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Bottom-up → ← **Top-down**

Atomic viewpoint:

- Explicitly consider discrete atomistic structure
- Solve for atomic trajectories and infer from these about material properties & behavior
- Features internal length scales (atomic distance)

"Many-particle system with statistical properties"

WEEK 3

Relevant scales in materials

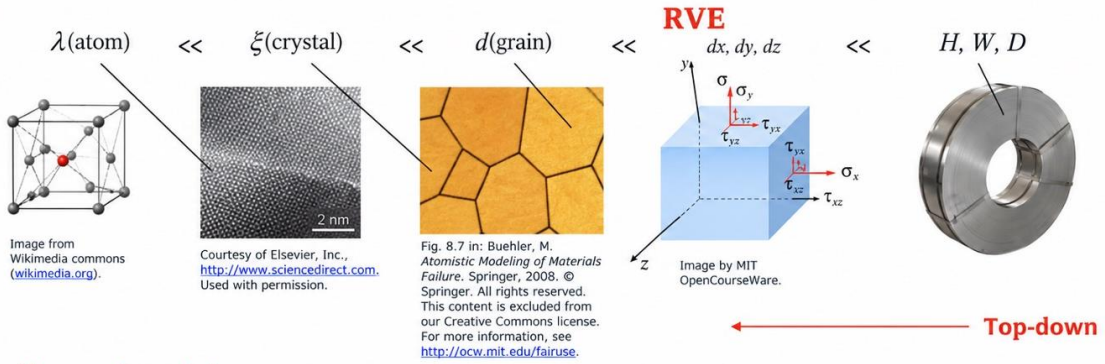


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Fig. 8.7 in: Buehler, M. *Atomistic Modeling of Materials Failure*. Springer, 2008. © Springer. All rights reserved. This content is excluded from our Creative Commons license. For more information, see http://ocw.mit.edu/fairuse.

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Continuum viewpoint:

- Treat material as matter with no internal structure.
- Develop mathematical model (governing equation) based on representative volume element (RVE), contains "enough" material such that internal structure can be neglected.
- Features no characteristic length scales, provided RVE is large enough.

"PDE with parameters"

WEEK 3

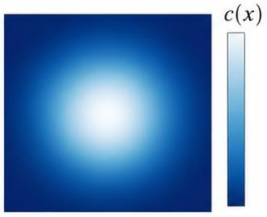
Goal of this section

Goal of this section

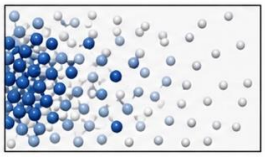
- **Diffusion** as example.
- **Continuum description** (top-down approach), partial differential equation.
- **Atomistic description** (bottom-up approach), based on dynamics of molecules, obtained via numerical simulation of the molecular dynamics.

Continuum description
(top-down approach)

$$\frac{\partial c(x,t)}{\partial t} = D\nabla^2 c(x,t)$$



Atomistic description
(bottom-up approach)



Goal of this section

Detailed Breakdown of the Continuum Formula (Top-Down Approach)

$$\frac{\partial c(x,t)}{\partial t} = D\nabla^2 c(x,t)$$

- $\frac{\partial c(x,t)}{\partial t}$ **(Rate of Change):** How concentration changes over time at a specific point.

- D **(Diffusion Coefficient / Diffusivity):** A constant measuring how fast particles diffuse through the medium (larger D means faster spreading).

- $\nabla^2 c(x,t)$ **(Laplacian of Concentration):** Measures the spatial curvature or concentration differences around a point.

    - In 1D, $\nabla^2 c = \frac{\partial^2 c}{\partial x^2}$ (the second spatial derivative).

    - **Physical meaning:** If a point is a "peak" (higher concentration than its neighbors), $\nabla^2 c$ is negative, causing concentration to decrease over time. If it is a "valley", concentration increases.

Brief Summary of Other Slide Content

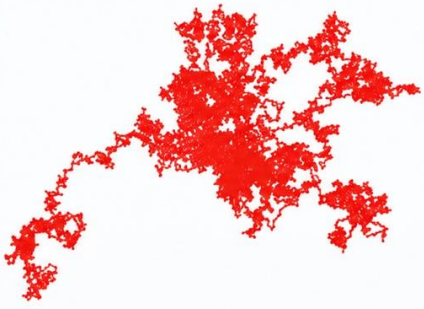
- **Section Goal:** Uses **diffusion** as the primary example to compare two modeling approaches.

- **Continuum Description (Top-Down):** Ignores individual atoms and models diffusion as a smooth field using a Partial Differential Equation (PDE).

- **Atomistic Description (Bottom-Up):** Models diffusion by simulating individual molecular movements using **Molecular Dynamics (MD)**.

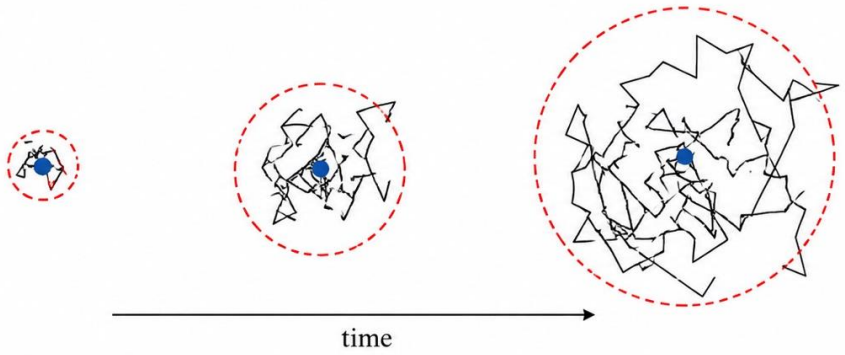
Microscopic mechanism: random walk or Brownian motion

- Brownian motion was first observed (1827) by the British botanist Robert Brown (1773-1858) when studying pollen grains in water.
- Initially thought to be sign of life, but later confirmed to be also present in inorganic particles.
- The effect was finally explained in 1905 by Albert Einstein, who realized it was caused by water molecules randomly smacking into the particles.



Public domain image.

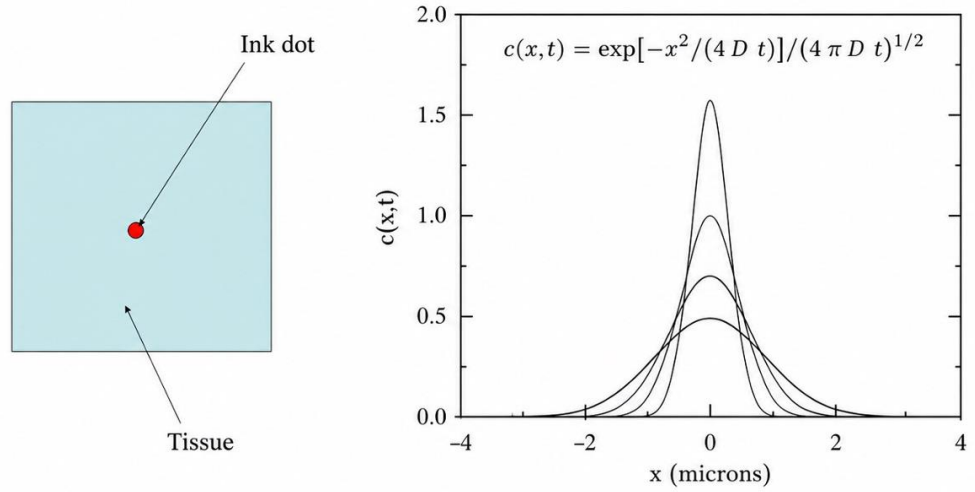
Brownian motion leads to net particle movement



Particle slowly moves away from its initial position.

WEEK 3

Macroscopic observation: concentration change



Slide 4 Breakdown: Macroscopic Observation (Concentration Change)

1. Physical Setup

- An **ink dot** is dropped at the center ($x = 0$) of a tissue at time $t = 0$.

- Over time, the sharp central spike flattens out and spreads wider across space.

2. The Analytical Solution Formula

The slide presents the 1D Gaussian analytical solution to the diffusion equation for a point source release:

$$c(x,t) = \frac{\exp\left(-\frac{x^2}{4Dt}\right)}{(4\pi Dt)^{1/2}}$$

- $x = 0$ **(Center):** As time t increases, the denominator $(4\pi Dt)^{1/2}$ increases, so the height of the central peak decreases.

- $x \neq 0$ **(Sides):** Over time, particles spread outward, causing the concentration in the surrounding region to rise.

- ## Quick Summary

Term	What it represents	Physical meaning
$c(x,t)$	**Concentration Field**	How much material is at spot x at time t.
$\frac{\partial c}{\partial t}$	**Rate of Change**	How fast the amount of material at spot x is changing right now.
$D\frac{\partial^2 c}{\partial x^2}$	**Diffusion Effect**	How spatial differences (curvature) drive the movement of material.

Area Under the Plot = Total Mass / Amount of Material

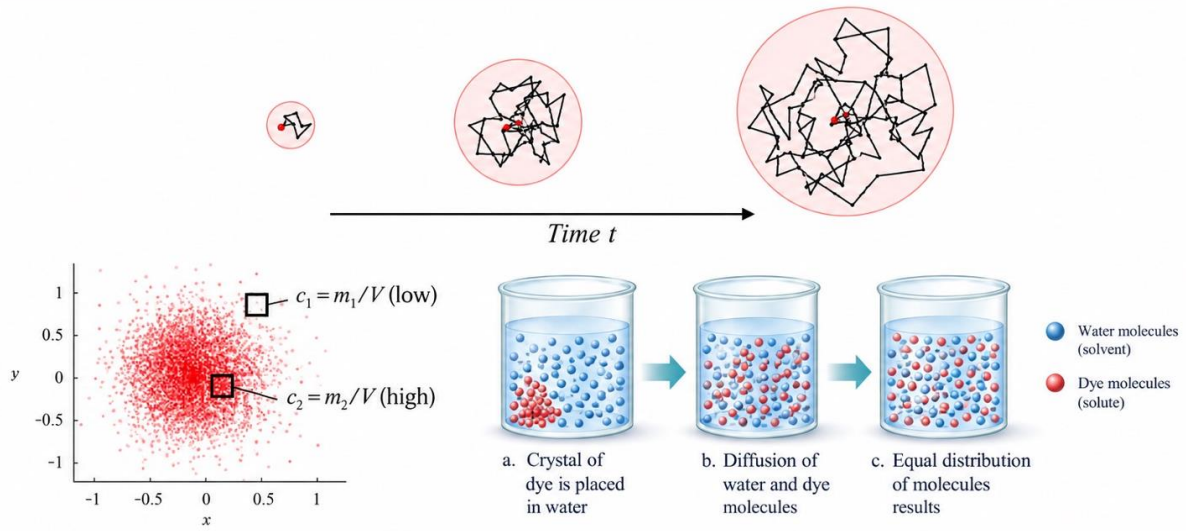
If you calculate the integral of concentration $c(x,t)$ across all space (x):

$$\int_{-\infty}^{\infty} c\,(x,t)\,dx = \text{Total Mass } (m)$$

- Because matter is conserved (no ink is created or destroyed as it diffuses), the **total area under the curve stays constant over time**, even as the peak flattens and spreads outward.

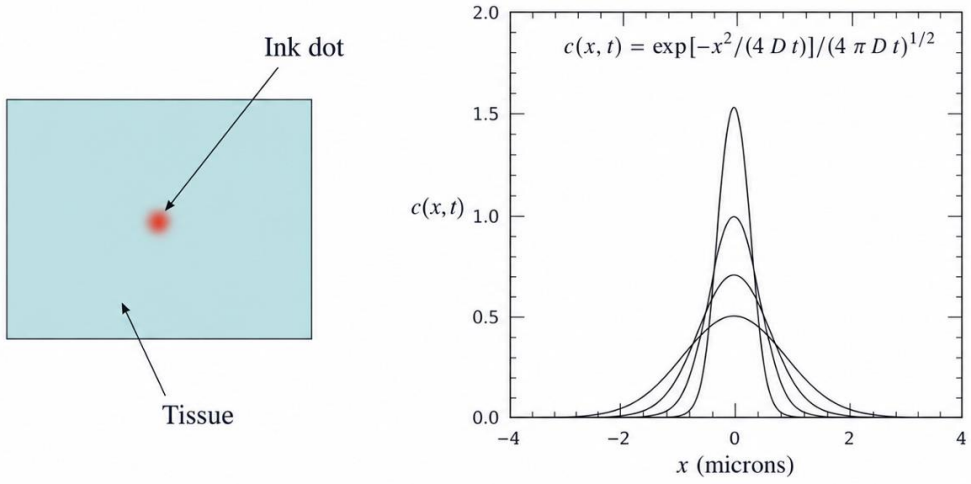
WEEK 3

Brownian motion leads to net particle movement

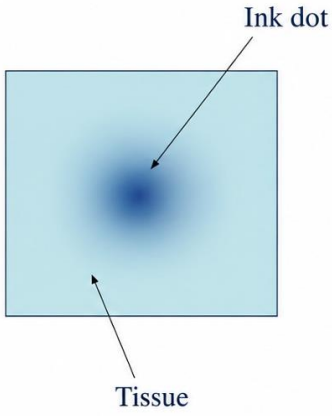


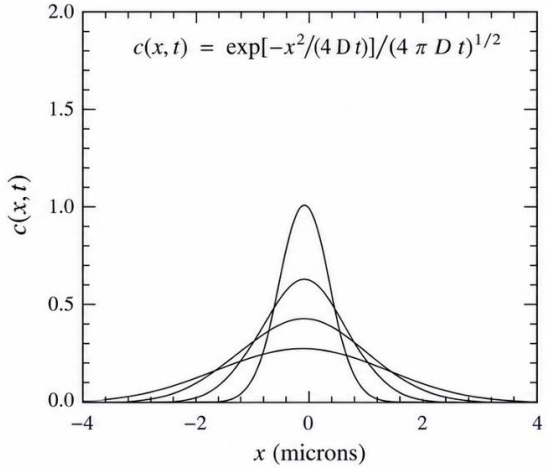
WEEK 3

Macroscopic observation: concentration change

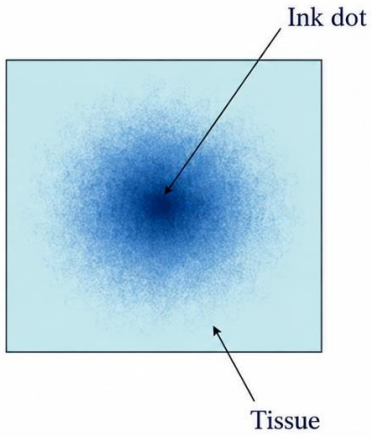


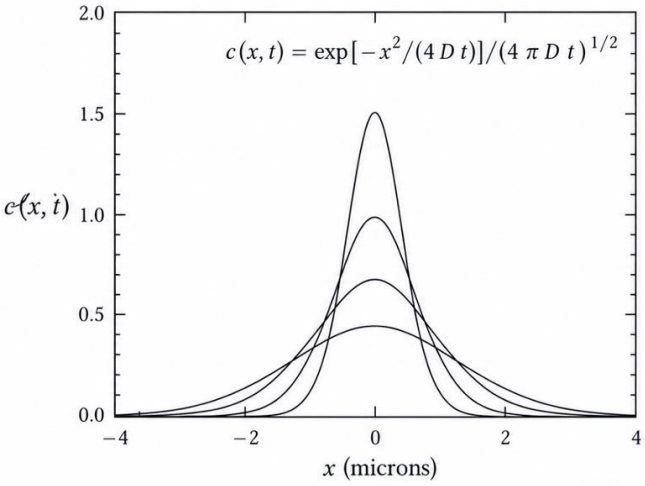
Macroscopic observation: concentration change





Macroscopic observation: concentration change

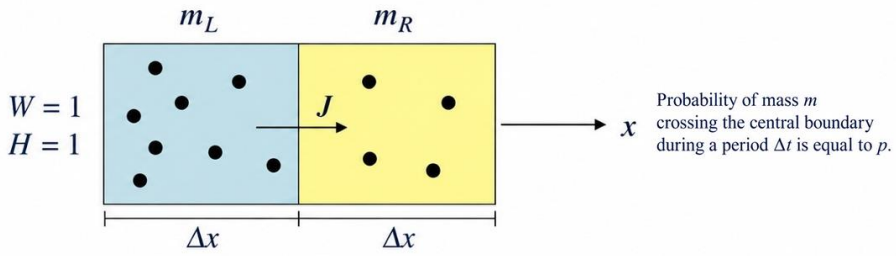




WEEK 3

Approach 1: Continuum model

- Develop differential equation based on differential element



J: Mass flux (mass per unit time per unit area)

Concept: Balance mass in a differential volume element; much greater in dimension than inhomogeneities ("sufficiently large REV")

Approach 1: Continuum Model

1. Core Goal & Setup

- **Goal:** Develop a continuous partial differential equation (PDE) by doing a mass balance over a small control volume (differential element).

- **Physical Setup:**
  - Two adjacent control volumes of width Δx, height $H = 1$, and width into page $W = 1$.
  - Left volume contains mass m_L, right volume contains mass m_R.
  - Particles cross the central boundary with a probability p over a time interval Δt.

2. Key Quantities Defined on Slide

- **Mass Flux (J):** Defined as **mass per unit time per unit area** passing across the boundary.

- Formula unit: $\frac{\text{mass}}{\text{time}\cdot\text{area}} = \frac{\text{kg}}{\text{s}\cdot\text{m}^2}$.

- J represents the net movement rate of mass from the left box to the right box.

- **Representative Elementary Volume (REV):**
  - To use a continuum model, the control volume ($\Delta x \cdot H \cdot W$) must be **sufficiently large** compared to micro-structural inhomogeneities (individual atomic gaps or local variations).
  - This allows us to smooth out individual particle bumps and define continuous properties like concentration $c(x, t)$.

3. Core Mass Balance Concept

- **Conservation of Mass:** The net mass moving across the interface per unit time creates the mass flux J:

$$J = \frac{\text{Net Mass Transferred}}{\text{Area} \cdot \Delta t} = \frac{p \cdot m_L - p \cdot m_R}{(H \cdot W) \cdot \Delta t}$$

- This simple discrete mass balance between two adjacent blocks is the starting step to derive **Fick's First Law** ($J = -D\frac{\partial c}{\partial x}$) and **Fick's Second Law** ($\frac{\partial c}{\partial t} = D\frac{\partial^2 c}{\partial x^2}$) in continuous space.

Must read First:

1. REV (Representative Elementary Volume)

The Problem: At a microscopic level, tissue or fluids are full of individual cells, molecules, and empty spaces. If you zoom in too far, "concentration" makes no sense because you are either pointing directly at one single atom or at empty space.

The Solution: An **REV** is simply the **smallest box you can draw** that contains *enough* particles so that properties like concentration, pressure, or density become smooth, stable averages.

- **Analogy:** Imagine looking at a digital photo. If you look at one single pixel, you just see a random red dot (microscopic scale). If you look at a box of 1000×1000 pixels, you can clearly see a human face (the REV/continuum scale).

- **Why it matters for modeling:** It allows us to stop counting individual molecules and start using smooth math equations (like calculus).

2. Mass Flux (J)

- **Plain Words:** Flux is simply **traffic flow for particles**.

- It measures: *"How much mass (m) passes through a specific doorway (area) in one second (time)?"*

- **Units:** $\mathrm{kg}/(\mathrm{m}^2 \cdot \mathrm{s})$.

3. Fick's First Law (Movement & Direction)

- **Equation:**

$$J = -D\frac{\partial c}{\partial x}$$

- **What it means in plain words:** Material naturally flows from high concentration to low concentration.
- **Breaking down the pieces:**

    - J **(Flux):** The rate and direction of particle flow.

    - D **(Diffusion Coefficient):** How easily the particle moves through that specific tissue/medium (e.g., water has a high D, thick mucus has a low D).

- $\frac{\partial c}{\partial x}$ **(Concentration Gradient / Slope):** How steeply the concentration drops over distance.

- **The Minus Sign ($-$)**: Particles move **downhill** (from high to low concentration). If concentration increases to the right (+ slope), flow goes to the left (- direction).

4. Fick's Second Law (Change Over Time)

- **Equation:**

$$\frac{\partial c}{\partial t} = D\frac{\partial^2 c}{\partial x^2}$$

- **What it means in plain words:** How fast a spot fills up or empties out over time depends on the **curvature** (concavity) of the concentration profile around it.

- **Breaking down the pieces:**

  - $\frac{\partial c}{\partial t}$**:** How fast concentration at a specific point goes up or down per second.

  - $\frac{\partial^2 c}{\partial x^2}$ **(Curvature / Second Derivative):**

    - If a point is a **peak** (sharply higher than its surroundings), it spreads out rapidly → concentration **drops** over time.

- If a point is a **valley** (sharply lower than its surroundings), material flows into it → concentration **rises** over time.

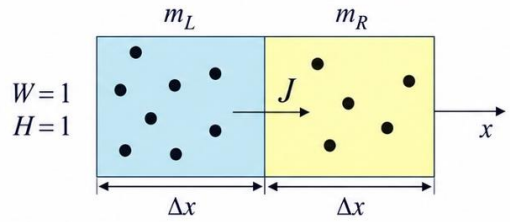
Summary Cheat Sheet for Your Course

Term	Simple Medical/Biomedical Analogy	Engineering/Mathematical Term
$c(x,t)$	Blood glucose level at a specific tissue spot right now	**Concentration Field** (Scalar)
J	Rate of drug molecules crossing a cell membrane per cm^2 per second	**Mass Flux Vector**
REV	Taking a tissue biopsy volume large enough to represent the whole organ	**Representative Elementary Volume**
Fick's 1st Law	Passive diffusion driven by partial pressure/gradient differences	**Flux Equation** ($J = -D\nabla c$)
Fick's 2nd Law	How a injected dye spot spreads out and flattens over time in tissue	**Diffusion Differential Equation** ($\frac{\partial c}{\partial t} = D\nabla^2 c$)

WEEK 3

Approach 1: Continuum model

- Develop differential equation based on differential element



Probability of mass m crossing the central boundary during a period Δt is equal to p.

$$J_L = \frac{1}{1 \times 1}\frac{p}{\Delta t} m_L \quad \text{Mass flux from left to right}$$

$$J_R = \frac{1}{1 \times 1}\frac{p}{\Delta t} m_R \quad \text{Mass flux from right to left}$$

Effective mass flux

$$J = \frac{1}{1 \times 1}\frac{p}{\Delta t}(m_L - m_R)$$

- *More mass, more flux (m_L is ~ to number of particles).*

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Slide 6: Approach 1: Continuum Model (Deriving Effective Mass Flux)

This slide takes the two adjacent boxes from the previous slide and writes down the exact step-by-step math to derive **Net Mass Flux (J)**.

1. Individual Flux Equations

- J_L **(Mass flux moving from Left to Right):**

$$J_L = \frac{1}{1 \times 1}\frac{p}{\Delta t} m_L$$

  - m_L**:** Total mass in the left box.
  - p**:** Probability that a particle crosses the middle line in time Δt.
  - $\frac{1}{1 \times 1}$**:** Division by the cross-sectional area (Area $= W \times H = 1 \times 1$).

- **Meaning:** This is the rate of mass per unit area leaving the left side and entering the right side.

- J_R **(Mass flux moving from Right to Left):**

$$J_R = \frac{1}{1 \times 1}\frac{p}{\Delta t}m_R$$

  - m_R**:** Total mass in the right box.

  - **Meaning:** Particles in the right box also randomly move back to the left box at the exact same probability rate $p/\Delta t$.

2. Effective (Net) Mass Flux (*J*)

Subtracting the two opposing flows gives the **net flux (*J*)** across the boundary:

$$J = J_L - J_R = \frac{1}{1 \times 1}\frac{p}{\Delta t}(m_L - m_R)$$

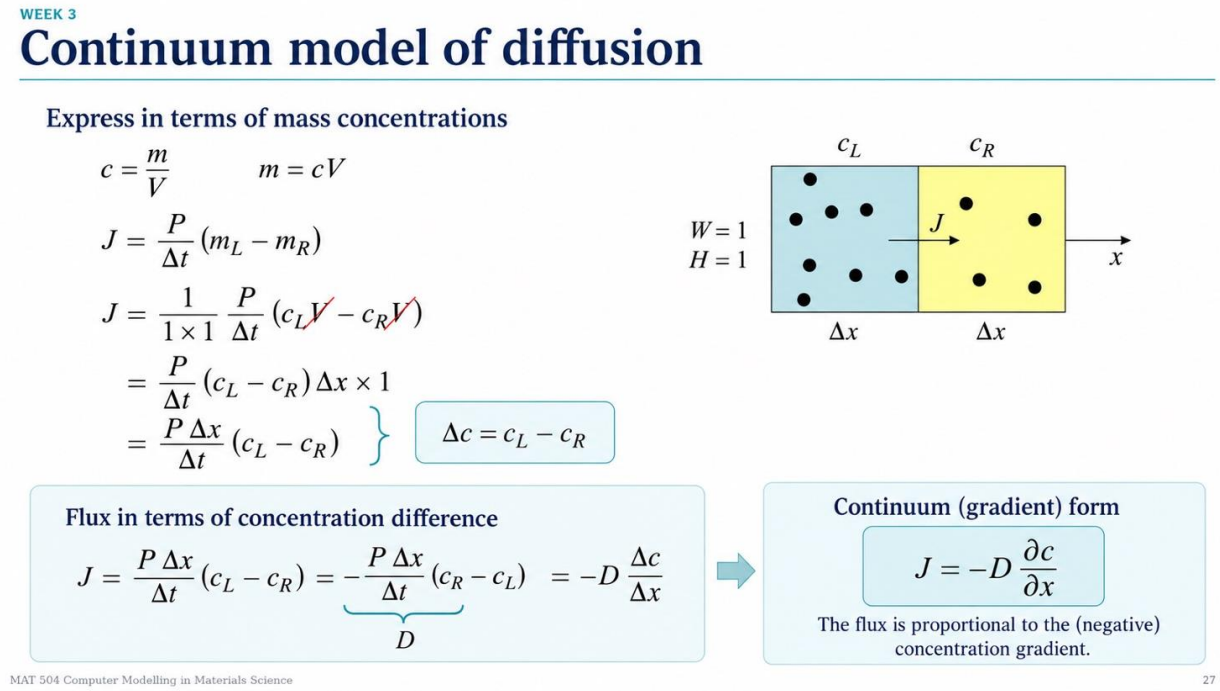
- **Key Takeaway:** Net movement (J) depends entirely on the **difference in mass** between the two sides ($m_L - m_R$).

- **"More mass, more flux":** Since mass is directly proportional to the number of particles ($m_L \propto N_L$), having more particles in the left box automatically leads to a higher net flow toward the right box.

Medical/Research Analogy

Think of two rooms connected by an open door. People in both rooms walk around randomly and cross the door with probability p.

- If Room L has 100 people and Room R has 20 people, more people cross from L to R than R to L.
- The **net traffic (J)** flows from L to R purely because of the difference in crowd size ($m_L - m_R$).



Slide 7: Continuum Model of Diffusion (Deriving Fick's First Law)

This slide connects discrete particle mass to continuous concentration, completing the mathematical derivation of **Fick's First Law**.

1. Substituting Mass with Concentration

- **General Definition:** Concentration is mass per unit volume ($c = m/V$), so mass is concentration times volume ($m = c \cdot V$).
- **Control Volume:** The volume of each box is $V = \Delta x \times H \times W = \Delta x \times 1 \times 1$.

- **Mass Expressions:**

  - $m_L = c_L \cdot (\Delta x \times 1 \times 1)$
  - $m_R = c_R \cdot (\Delta x \times 1 \times 1)$

Substitute m_L and m_R into the flux equation:

$$J = \frac{p}{\Delta t}(m_L - m_R) = \frac{p}{\Delta t}(c_L \cdot \Delta x - c_R \cdot \Delta x) = \frac{p\Delta x}{\Delta t}(c_L - c_R)$$

2. Defining Diffusivity (D) & Spatial Differences

- **Flipping terms:** To match the standard direction of change (Right minus Left: $c_R - c_L = \Delta c$), we factor out a negative sign:

$$c_L - c_R = -(c_R - c_L) = -\Delta c$$

- **Defining D (Diffusion Coefficient):** The microscopic jump parameters are grouped into a single macroscopic constant:

$$D = \frac{p(\Delta x)^2}{\Delta t}$$

(Note: Expanding $p\Delta x/\Delta t$ by multiplying numerator and denominator by Δx defines D).

This yields the discrete concentration flux equation:

$$J = -D\frac{\Delta c}{\Delta x}$$

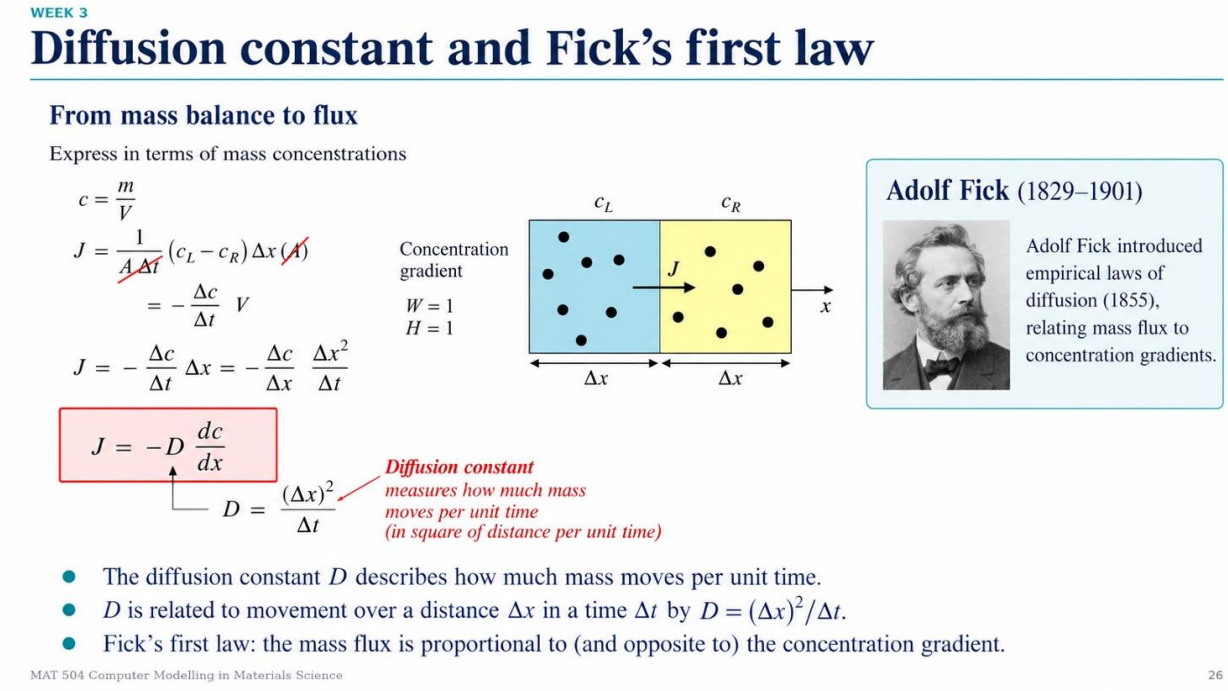
3. Transition to Continuous Form (Fick's First Law)

By taking the limit as spatial box width shrinks to zero ($\Delta x \to 0$), discrete differences ($\frac{\Delta c}{\Delta x}$) turn into a continuous derivative ($\frac{\partial c}{\partial x}$):

$$J = -D\frac{\partial c}{\partial x}$$

- **Physical Meaning:** The mass flux J is directly proportional to the **negative concentration gradient**.

- **The Minus Sign ($-$):** Ensures material flows **downhill** from high concentration to low concentration.



1. Algebraic Bridge (Micro to Macro)

The slide re-arranges the mass balance to explicitly separate the microscopic geometry ($\Delta x, \Delta t$) from the concentration spatial difference ($\frac{\Delta c}{\Delta x}$):

$$J = -\frac{\Delta c}{\Delta t}\Delta x = -\frac{\Delta c}{\Delta x}\frac{(\Delta x)^2}{\Delta t} = -D\frac{\partial c}{\partial x}$$

2. Physical Definition of Diffusivity (D)

- **Formula:** $D = \frac{(\Delta x)^2}{\Delta t}$.

- **Physical Meaning:** Measures how much area/mass is covered by random microscopic jumps per unit time (units: m^2/s).

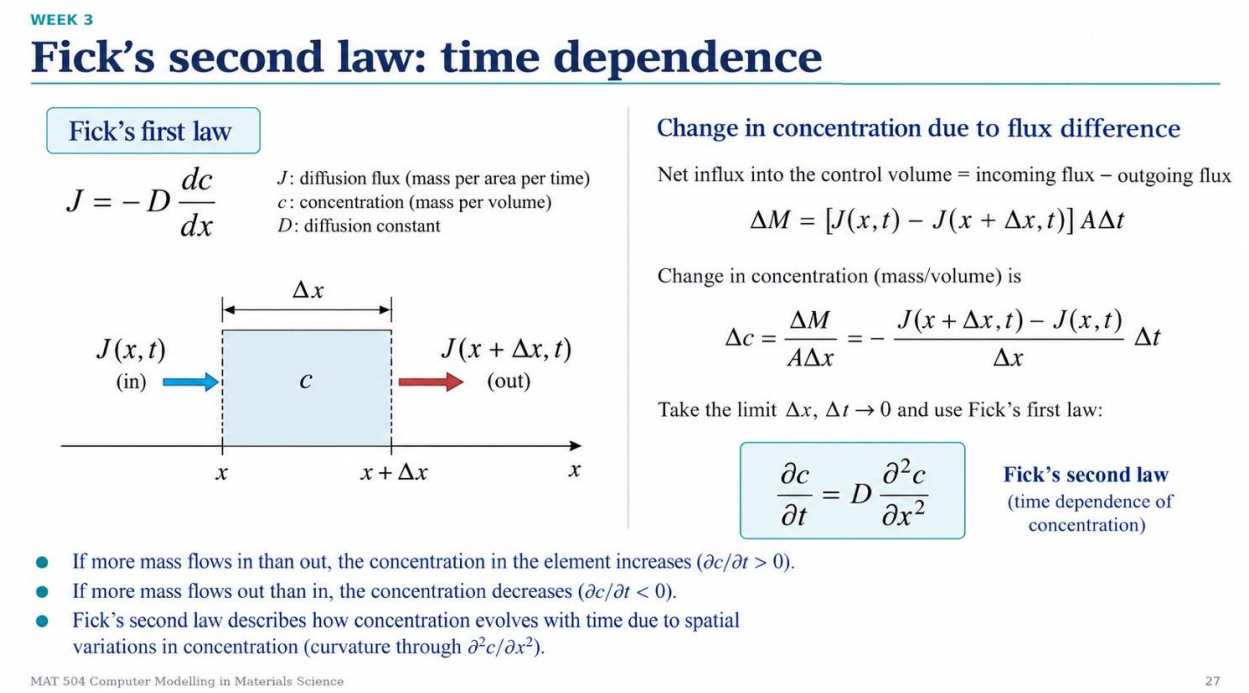
- **Microscopic Link:** Embeds the atomic hopping distance (Δx) and time scale (Δt) into a single continuum coefficient D.

3. Key Takeaways from the Slide

- **Diffusion Constant (D):** Quantifies mass movement per unit time over a given step distance squared.

- **Fick's First Law ($J = -D\frac{dc}{dx}$):** Mass flux is directly proportional to—and flows in the opposite direction of—the concentration gradient.

- **Historical Context:** Adolf Fick (1855) first formulated these empirical laws to link macroscopic flux directly to concentration slopes.



While Fick's 1st Law defines traffic flow (J), this slide derives **Fick's 2nd Law**, which calculates how local concentration (c) changes over time (Δt) based on net mass accumulation.

1. Mass Balance Over a Single Control Element

- **Setup:** A single block of width Δx and cross-sectional area A.
- **Mass In vs. Mass Out:**
    - Mass entering at position x: $J(x,t) \cdot A \cdot \Delta t$
    - Mass leaving at position $x + \Delta x$: $J(x+\Delta x,t) \cdot A \cdot \Delta t$

- $J(x,t) \cdot A \cdot \Delta t$ **= Mass entering the box**
    - It takes the mass flow rate per area (J), multiplies it by the wall's surface area (A), and multiplies by the elapsed time (Δt) to give the total kilograms/moles entering at the left boundary (x).

- $J(x + \Delta x, t) \cdot A \cdot \Delta t$ **= Mass leaving the box**
  - Same calculation (Flux × Area × Time), but measured at the right boundary ($x + \Delta x$) to calculate the total mass exiting the box.
    -

- **Net Accumulation (ΔM):**

$$\Delta M = [J(x,t) - J(x + \Delta x, t)]A\Delta t$$

Net accumulation means **how much mass gets trapped inside the box** over time.

- **Formula:** Mass In $-$ Mass Out $= \Delta M$
- **If positive ($\Delta M > 0$):** More mass enters than leaves → Concentration **increases**.
- **If negative ($\Delta M < 0$):** More mass leaves than enters → Concentration **decreases**.

2. From Mass Accumulation to Concentration Change

To get concentration change ($\Delta c = \frac{\Delta M}{\text{Volume}}$ where Volume $= A\Delta x$):

$$\Delta c = \frac{\Delta M}{A\Delta x} = -\frac{J(x + \Delta x, t) - J(x,t)}{\Delta x}\Delta t$$

Divide both sides by Δt and take the continuum limit ($\Delta x, \Delta t \to 0$):

$$\frac{\partial c}{\partial t} = -\frac{\partial J}{\partial x}$$

- **Physical Meaning:** The rate of concentration change over time equals the **negative spatial change in flux**.

3. Deriving Fick's Second Law

Substitute **Fick's 1st Law** ($J = -D\frac{\partial c}{\partial x}$) into the expression above:

$$\frac{\partial c}{\partial t} = -\frac{\partial}{\partial x}\left(-D\frac{\partial c}{\partial x}\right) = D\frac{\partial^2 c}{\partial x^2}$$

4. Summary & Intuition

- **If** Flux In $>$ Flux Out**:** Mass accumulates, so concentration rises ($\frac{\partial c}{\partial t} > 0$).

- **If** Flux Out $>$ Flux In**:** Mass depletes, so concentration drops ($\frac{\partial c}{\partial t} < 0$).

- **Core Takeaway:** Fick's Second Law describes how concentration evolves over time due to spatial variations in concentration—specifically driven by the local **curvature** ($\frac{\partial^2 c}{\partial x^2}$) of the profile.

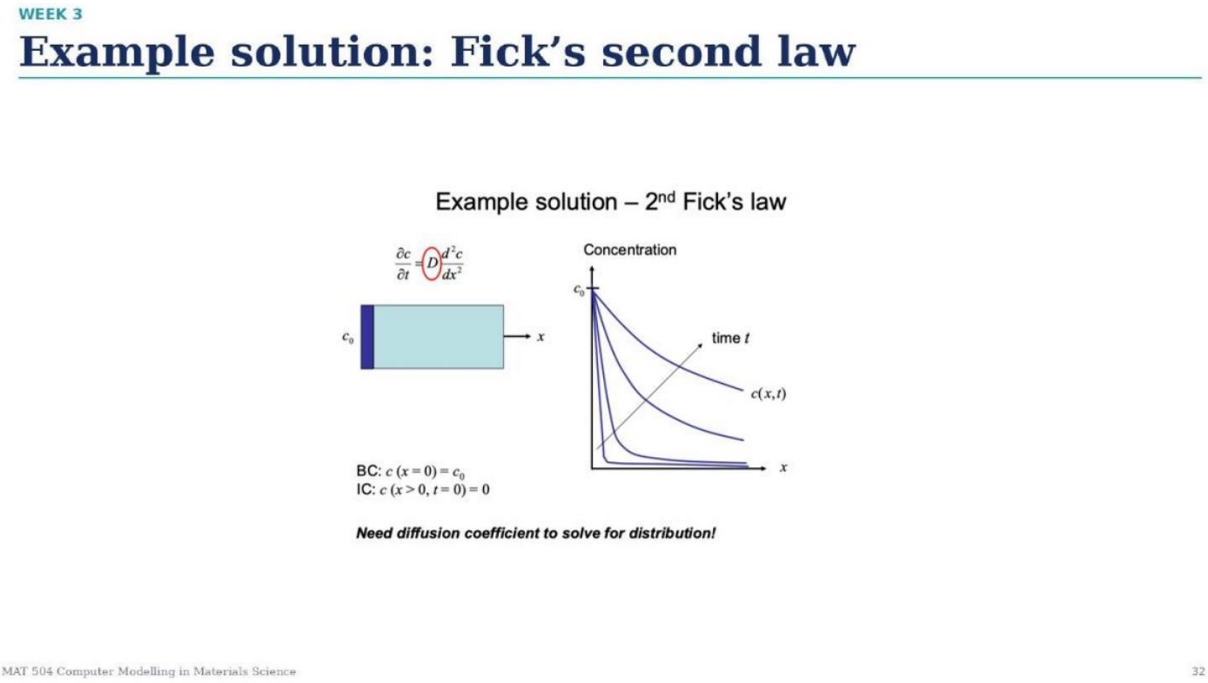
- **Interpretation:** Fick's Second Law shows you precisely *where* the tissue is gaining drug (the edges) and *where* it is losing drug (the center) at any given second.

  Spot on! You hit the exact intuitive distinction between the two laws:

- **Fick's 1st Law ($J = -D\frac{\partial c}{\partial x}$):** Answers *"How fast and in what direction is mass flowing across a boundary right now?"* It calculates the **traffic rate (flux J)** across space based on the concentration slope.

- **Fick's 2nd Law ($\frac{\partial c}{\partial t} = D\frac{\partial^2 c}{\partial x^2}$):** Answers *"How fast is the concentration filling up or draining at one specific location over time?"* It calculates the **rate of accumulation ($\frac{\partial c}{\partial t}$)** based on the difference between traffic coming in versus traffic going out (the local curvature).

In simple terms: **1st Law = Movement/Traffic**, **2nd Law = Accumulation/Time Change**.



This slide illustrates what happens when you apply a continuous, endless supply of a substance to the edge of an empty material and watch it soak in over time.

1. The Setup (Initial and Boundary Conditions)

To solve any differential equation like Fick's Second Law, you must define the starting rules of your specific scenario.

- **Initial Condition (IC):** $c(x > 0, t = 0) = 0$. This simply means that at time zero ($t = 0$), before the process starts, the entire block of material ($x > 0$) is completely empty (concentration is zero).

- **Boundary Condition (BC):** $c(x = 0) = c_0$. This means the very left edge of the material ($x = 0$) is permanently locked at a high concentration (c_0).

- **Medical Analogy:** Think of applying a fentanyl patch to the skin. The patch maintains a constant, high concentration of drug right at the skin surface (c_0), while the tissue underneath starts completely empty (zero).

2. The Graph (Concentration Profiles Over Time)

The plot visually maps the mathematical solution to Fick's Second Law for this exact scenario.

- **Y-axis (Concentration) and X-axis (Distance):** Every curve shows how much mass has penetrated into the material at a specific snapshot in time.

- **The Left Anchor:** Notice how every single curve starts at exactly c_0 on the y-axis. This is the Boundary Condition in action; the edge never runs out of mass.

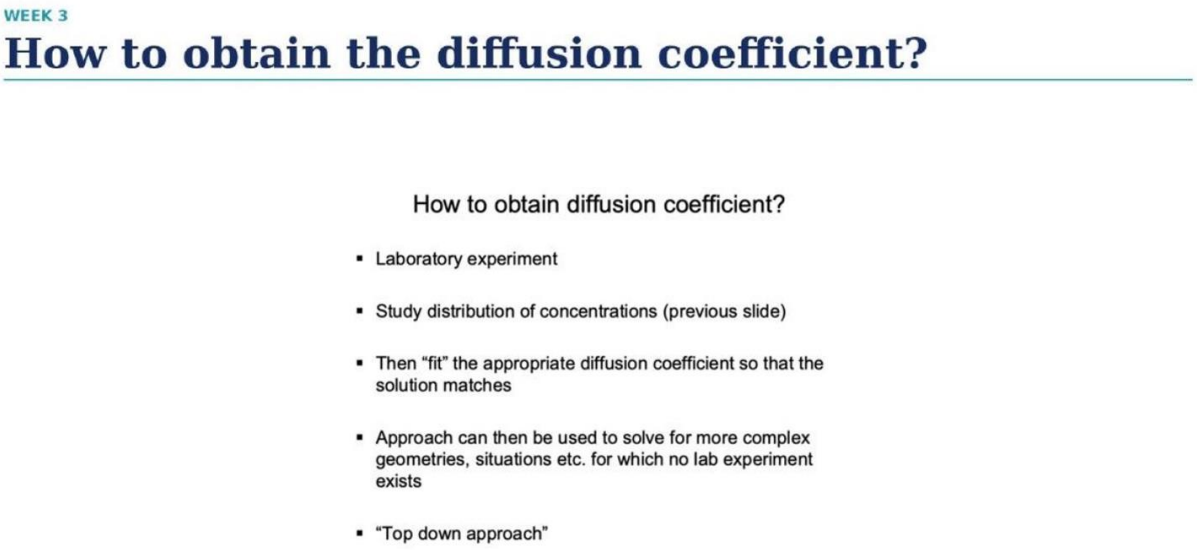
- **The Spreading (Time t Arrow):**
    - The steep, L-shaped curve closest to the axes represents a very early time. The mass has barely penetrated the material, dropping rapidly to zero.
    - As time increases (following the "time t" arrow), the curves stretch further to the right. The mass is diffusing deeper into the material, raising the concentration further away from the source.

3. The Role of the Diffusion Coefficient (D)

- The equation $\frac{\partial c}{\partial t} = D\frac{\partial^2 c}{\partial x^2}$ governs this entire spreading process.

- The slide emphasizes that you **"Need diffusion coefficient to solve for distribution!"**. While the math tells you the *shape* of those curves, D dictates exactly *how fast* those

curves sweep to the right. A high D (like gas in air) means the curves will flatten out in seconds; a low D (like a large drug molecule in dense tissue) means it could take days to reach the same profile.



Slide Breakdown: How to Obtain the Diffusion Coefficient?

This slide answers the practical engineering question: **How do we actually get the value of D to use in our math models?**

It presents the **Top-Down Approach** (experimental curve fitting).

1. Step-by-Step Experimental Method

1. **Run a Laboratory Experiment:** Set up a simple physical sample in the lab (like the one-dimensional tissue/material block from the previous slide).

2. **Measure Concentration Profiles:** Track the real concentration of the substance at different distances x and different times t (e.g., using imaging, fluorescent markers, or chemical sampling).

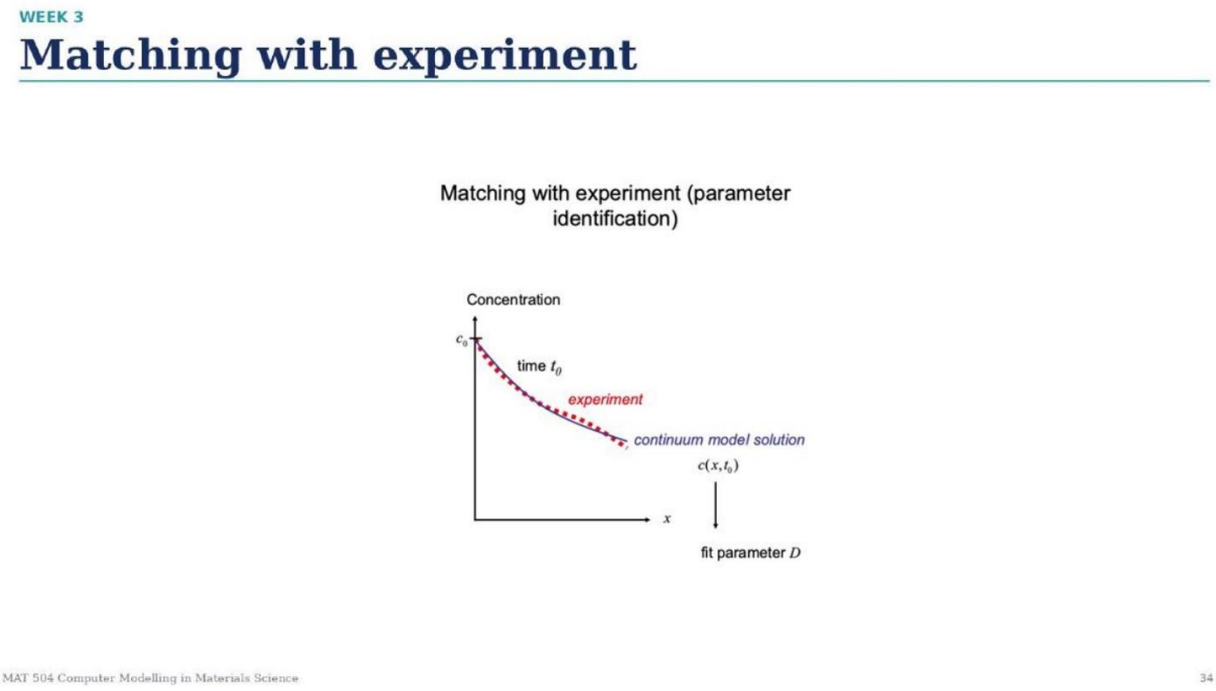
3. **Curve Fitting:** Take your analytical/mathematical curves from Fick's 2nd Law solution and adjust the single parameter D until the mathematical curves overlay perfectly onto your actual lab data points.

2. Why Do We Do This? (The Engineering Goal)

- **Extrapolation to Complex Problems:** You cannot build a full physical lab experiment for every complex 3D organ, drug delivery system, or material shape.
- Once you extract the true material property D from a simple lab setup, you can plug that exact same D into computer simulations to solve for **complex geometries and real-world clinical situations** where experiments are impossible.

3. Why It's Called a "Top-Down Approach"

- **Top-Down:** You observe **macroscopic lab behavior** (the overall spreading of mass) and work backward to extract the bulk coefficient D without needing to know atomic details.
- **Contrast (Bottom-Up):** Calculating D from microscopic atomic physics ($D = \frac{(\Delta x)^2}{\Delta t}$ or jump probabilities p).



Slide Breakdown: Matching with Experiment (Parameter Identification)

This slide visually details how the curve-fitting process works to extract the unknown diffusion coefficient D.

1. What the Plot Shows

- **Red Dotted Line (Experiment):** Data points measured in a laboratory at a specific time snapshot t_0.

- **Blue Solid Line (Continuum Model Solution):** The theoretical concentration curve $c(x, t_0)$ calculated from solving Fick's Second Law.

2. Parameter Identification Process

1. **Fix Time (t_0):** You take experimental concentration measurements across distance x at a fixed time t_0.

2. **Overlay the Math Model:** You plot your mathematical solution $c(x, t_0)$ on top of the experimental points.

3. **Adjust D:** You tune or "fit" the single unknown parameter D in the mathematical equation until the blue curve overlays the red experimental dots.

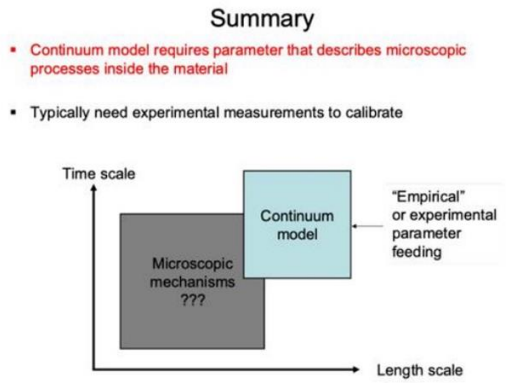
3. Core Purpose

- **Parameter Extraction:** This process identifies the true material property D for a given substance and medium.

- Once D is matched to this baseline experiment, it can be used in full-scale computer simulations.

WEEK 3

Summary



Slide Breakdown: Week 3 Summary

This slide ties together the entire philosophy of bridging scales in computer modeling.

1. The Core Modeling Dilemma

- **Scale Disconnect: Continuum models** operate at larger **length scales** (mm to meters) and longer **time scales** (seconds to hours), whereas **microscopic mechanisms** happen at atomic length scales (nm) and rapid time scales (ps).
- **The Bridge Problem:** To run a continuum model (like Fick's Second Law) without simulating every single atom, the model requires a single parameter—the **diffusion coefficient (D)**—that encapsulates all the underlying microscopic hopping behavior.

2. Key Bullet Takeaways

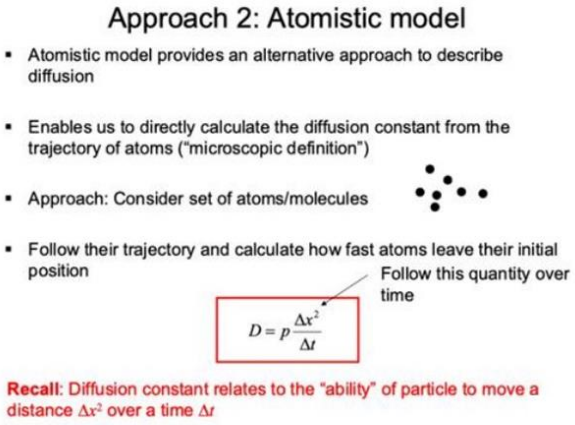
- **Microscopic Integration:** A continuum model needs a parameter that describes internal microscopic processes.
- **Experimental Calibration:** Because tracking trillions of microscopic mechanisms directly is often impossible or computationally prohibitive, we typically rely on **experimental measurements** to calibrate (fit) that continuum parameter (D).
- **"Empirical" Parameter Feeding:** Lab experiments feed empirical macroscopic parameters into continuum models, bypassing the need to solve every microscopic step directly.

3. Big-Picture Overview of the Module

- **Bottom-Up:** We proved mathematically that discrete microscopic hopping leads to bulk continuum laws ($D = \frac{p(\Delta x)^2}{\Delta t}$).
- **Top-Down:** In practice, engineers use lab curve-fitting to extract D directly from macroscopic experimental data.
- **Application:** Once D is calibrated, it feeds directly into computer simulations to model complex real-world geometries (e.g., drug delivery, tissue transport).

WEEK 3

Approach 2: Atomistic model



Slide Breakdown: Approach 2: Atomistic Model

While the previous approach measured D using macroscopic lab curves ("top-down"), this slide introduces the alternative **"bottom-up" atomistic modeling approach**.

1. Core Concept

- **Alternative Route:** The atomistic model provides a direct way to calculate the diffusion constant from the actual microscopic trajectories of individual atoms or molecules.
- **Microscopic Definition:** Instead of measuring bulk concentration profiles, you track discrete particle movements over time.

2. The Method (Tracking Trajectories)

1. **Consider a set of atoms/molecules:** Simulate or track a collection of individual particles.
2. **Follow their trajectories:** Measure how far particles travel away from their initial starting positions as time passes.
3. **Calculate D:** Relate particle displacement to time using the microscopic definition:

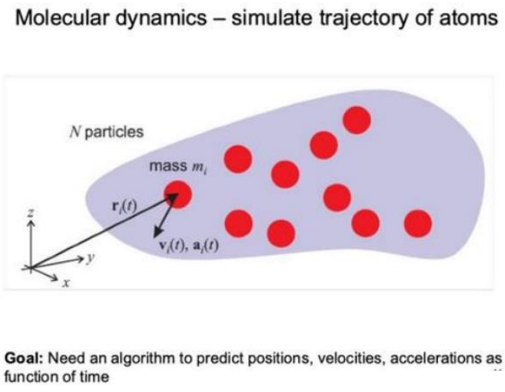
$$D = p\frac{\Delta x^2}{\Delta t}$$

3. Key Takeaway

- **Physical Interpretation:** The diffusion constant D quantifies the "ability" of a particle to displace over a step distance squared (Δx^2) within a given timeframe (Δt) weighted by jump probability p.

WEEK 3

Molecular dynamics: simulate trajectories of atoms



Slide Breakdown: Molecular Dynamics: Simulate Trajectories of Atoms

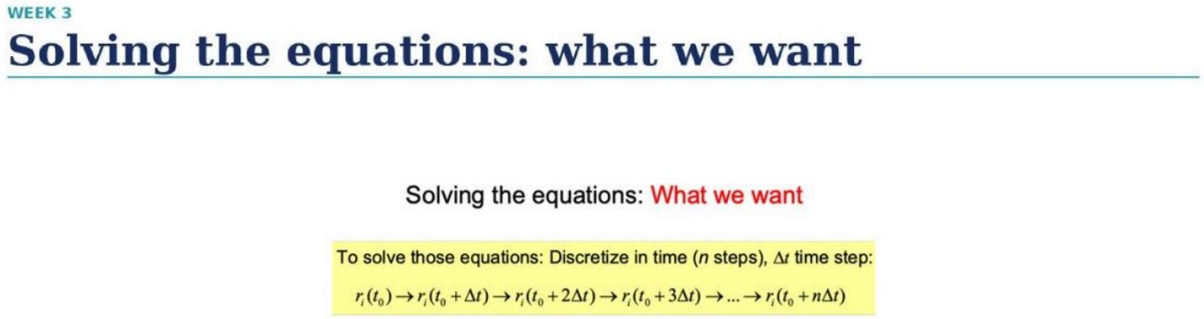
This slide shows how computational models implement the bottom-up atomistic approach by simulating individual particle trajectories using physics.

1. Core Concept of Molecular Dynamics (MD)

- **What it is:** Molecular Dynamics is a computational technique that simulates the physical movement of N discrete particles (atoms or molecules) in a system over time.
- **Setup:** Each particle i has an assigned mass m_i, position vector $\mathbf{r}_i(t)$, velocity $\mathbf{v}_i(t)$, and acceleration $\mathbf{a}_i(t)$.

2. The Computational Goal

- **Goal:** You need a numerical algorithm to iteratively calculate and predict the new positions, velocities, and accelerations of every single particle as a function of time.
- **How it connects to Diffusion:** By tracking these position trajectories $\mathbf{r}_i(t)$ over time, you calculate how far particles travel, allowing you to compute D directly from atom-scale physics.



Slide Breakdown: Solving the Equations: What We Want

This slide explains the practical core of **Molecular Dynamics (MD)**: how a computer actually tracks particle movement across time.

1. Time Discretization (The "Flipbook" Concept)

- **Continuous vs. Discrete:** Real physical time flows continuously, but a computer cannot process infinite smooth time.

- **Time Step (Δt):** The computer breaks time down into tiny, fixed intervals called time steps (Δt).

- n **Steps:** The simulation moves forward step-by-step over n total iterations. Think of it like a movie made of individual frames spaced Δt apart.

2. The Position Trajectory Chain

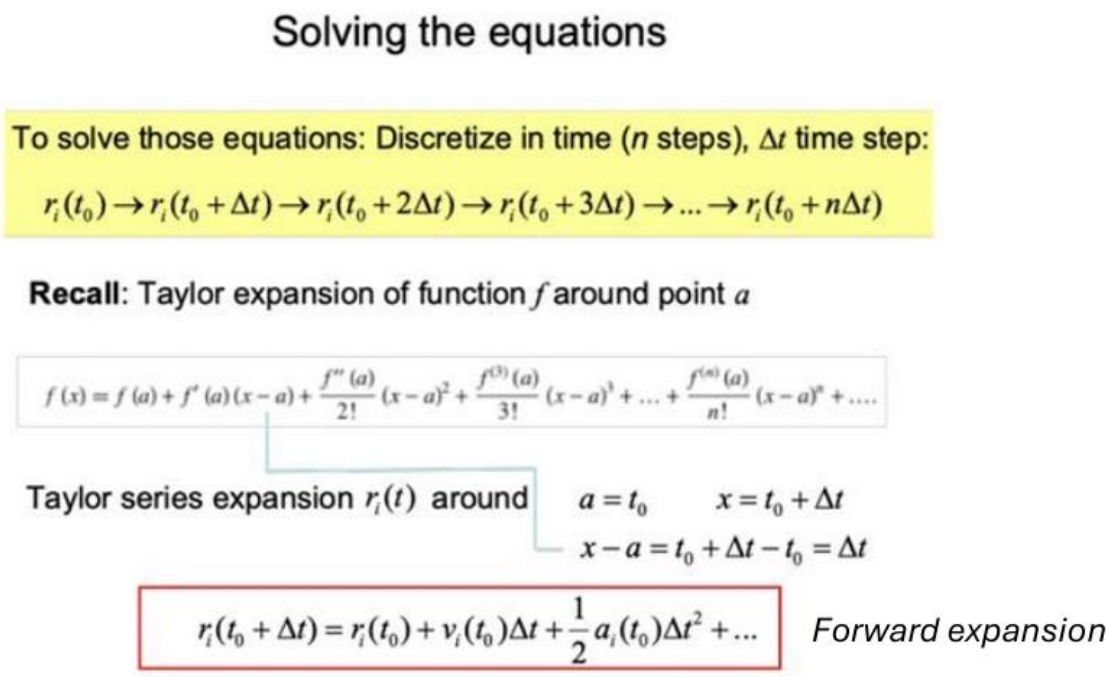
The highlighted formula shows the step-by-step evolution of particle i's position:

$$r_i(t_0) \rightarrow r_i(t_0 + \Delta t) \rightarrow r_i(t_0 + 2\Delta t) \rightarrow r_i(t_0 + 3\Delta t) \rightarrow \cdots \rightarrow r_i(t_0 + n\Delta t)$$

- $r_i(t_0)$**:** Starting position of particle i at time zero.
- $r_i(t_0 + \Delta t)$**:** Position after 1 time step.
- $r_i(t_0 + n\Delta t)$**:** Final position after n steps.

3. Medical & Physical Intuition

- **Goal:** By recording this exact trajectory path over thousands of steps, you can measure how far particles drift from their starting points.
- **Connecting to** D**:** Once you have the full trajectory chain, you calculate the average distance squared that particles traveled (Δx^2) over total time ($n\Delta t$), directly yielding the microscopic diffusion coefficient $D = \frac{\Delta x^2}{\Delta t}$.



Slide Breakdown: Solving the Equations (Taylor Series Expansion)

This slide bridges pure calculus and physical simulation by showing how a computer uses a **Taylor Series expansion** to calculate a particle's future position $r_i(t_0 + \Delta t)$.

1. The Mathematical Tool: Standard Taylor Series

- **General Calculus Definition:** A Taylor series predicts the value of a function $f(x)$ near a known point a using its derivatives:

$$f(x) = f(a) + f'(a)(x-a) + \frac{f''(a)}{2!}(x-a)^2 + \frac{f^{(3)}(a)}{3!}(x-a)^3 + \cdots$$

2. Mapping Calculus to Molecular Dynamics

To predict where a particle moves in the next time step, we map the generic calculus variables to real physical quantities:

- **Known point (a):** Current time t_0.

- **Target point (x):** Next time step $t_0 + \Delta t$.

- **Time step interval ($x - a$):** $(t_0 + \Delta t) - t_0 = \mathbf{\Delta t}$.

- **Function $f(t)$:** Position vector $r_i(t)$.

- **1st Derivative $f'(t)$:** Velocity vector $v_i(t)$.

- **2nd Derivative $f''(t)$:** Acceleration vector $a_i(t)$.

3. The Result: Forward Expansion Formula

Plugging these terms into the Taylor series yields the red-boxed equation for **Forward Expansion**:

$$r_i(t_0 + \Delta t) = r_i(t_0) + v_i(t_0)\Delta t + \frac{1}{2}a_i(t_0)\Delta t^2 + \cdots$$

4. Physical Intuition

This formula is simply high school kinematics (like $x = x_0 + vt + \frac{1}{2}at^2$) derived directly from fundamental calculus. It tells the computer:

1. Start at the current position $r_i(t_0)$.

2. Move forward based on current velocity $v_i(t_0)\Delta t$.

3. Adjust for current acceleration $\frac{1}{2}a_i(t_0)\Delta t^2$ caused by forces acting on the particle.

By repeatedly running this forward step, the computer steps particle i forward through time!

Because a computer cannot calculate infinite terms, it truncates (cuts off) the Taylor series after the first few terms!

In computational physics and Molecular Dynamics, this truncation is handled through **numerical integration algorithms** (most famously, the **Verlet Algorithm**).

1. Truncation and Numerical Error

When we cut the series off after the Δt^2 term, we throw away the higher-order terms ($\Delta t^3, \Delta t^4, \ldots$):

$$r_i(t_0 + \Delta t) \approx r_i(t_0) + v_i(t_0)\Delta t + \frac{1}{2}a_i(t_0)\Delta t^2 + \mathcal{O}(\Delta t^3)$$

- **The Trade-Off:** The terms we throw away are called the **Truncation Error** ($\mathcal{O}(\Delta t^3)$).

- **Why it works:** Because the time step Δt is chosen to be extremely small (e.g., femtoseconds, 10^{-15} s), Δt^3 becomes practically zero (10^{-45}), making the discarded terms negligible!

2. The Clever Trick: Verlet Algorithm (Canceling Velocity)

To avoid calculating velocities continuously and to make the math even more accurate, computers combine a **Forward Expansion** and a **Backward Expansion**:

1. **Forward Step:** $r(t + \Delta t) = r(t) + v(t)\Delta t + \frac{1}{2}a(t)\Delta t^2 + \frac{1}{6}b(t)\Delta t^3 + \mathcal{O}(\Delta t^4)$

2. **Backward Step:** $r(t-\Delta t) = r(t) - v(t)\Delta t + \frac{1}{2}a(t)\Delta t^2 - \frac{1}{6}b(t)\Delta t^3 + \mathcal{O}(\Delta t^4)$

Adding these two equations together causes the velocity terms ($v(t)\Delta t$) and the odd-power error terms (Δt^3) to **cancel out completely**:

$$r(t+\Delta t) = 2r(t) - r(t-\Delta t) + a(t)\Delta t^2 + \mathcal{O}(\Delta t^4)$$

3. How the Computer Runs It Step-by-Step

With this truncated formula, the computer calculates the future position using only simple arithmetic:

1. **Force & Acceleration:** Calculate current acceleration $a(t)$ using Newton's Second Law ($F = ma \rightarrow a = \frac{F}{m}$, where forces come from atomic interaction potentials).

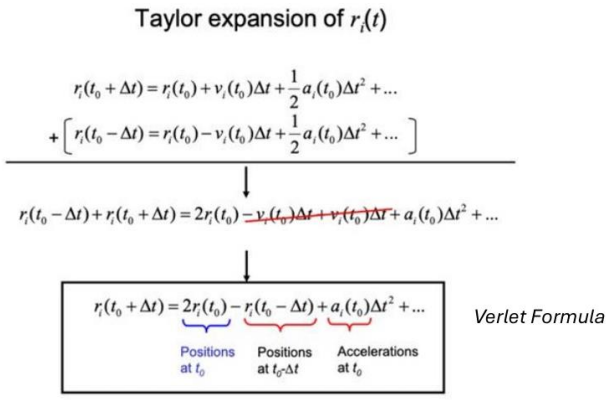
2. **Next Position:** Plug $r(t)$, previous position $r(t-\Delta t)$, and $a(t)$ into the formula to find $r(t+\Delta t)$.

3. **Repeat:** Advance time by Δt and repeat for n steps!

Would you like to look at how forces F are calculated between atoms, or jump straight to computing D from these trajectories?

WEEK 3

Taylor expansion of x(t)



Slide Breakdown: Taylor Expansion of $x(t)$ / Verlet Formula

This slide derives the **Verlet Algorithm**, demonstrating how to bypass velocities entirely to compute particle positions over time.

1. The Math: Adding Forward and Backward Expansions

- **Forward Step:** $r_i(t_0 + \Delta t) = r_i(t_0) + v_i(t_0)\Delta t + \frac{1}{2}a_i(t_0)\Delta t^2 + \cdots$
- **Backward Step:** $r_i(t_0 - \Delta t) = r_i(t_0) - v_i(t_0)\Delta t + \frac{1}{2}a_i(t_0)\Delta t^2 + \cdots$
- **Summing Both:** Adding the two equations together eliminates the velocity terms ($-v_i(t_0)\Delta t + v_i(t_0)\Delta t = 0$), leaving:

$$r_i(t_0 - \Delta t) + r_i(t_0 + \Delta t) = 2r_i(t_0) + a_i(t_0)\Delta t^2 + \cdots$$

2. The Final Verlet Formula

Rearranging the sum isolates the future position $r_i(t_0 + \Delta t)$:

$$r_i(t_0 + \Delta t) = 2r_i(t_0) - r_i(t_0 - \Delta t) + a_i(t_0)\Delta t^2 + \cdots$$

The three color-coded components of this equation are:

- $2r_i(t_0)$ **(Blue):** The current position at time t_0.

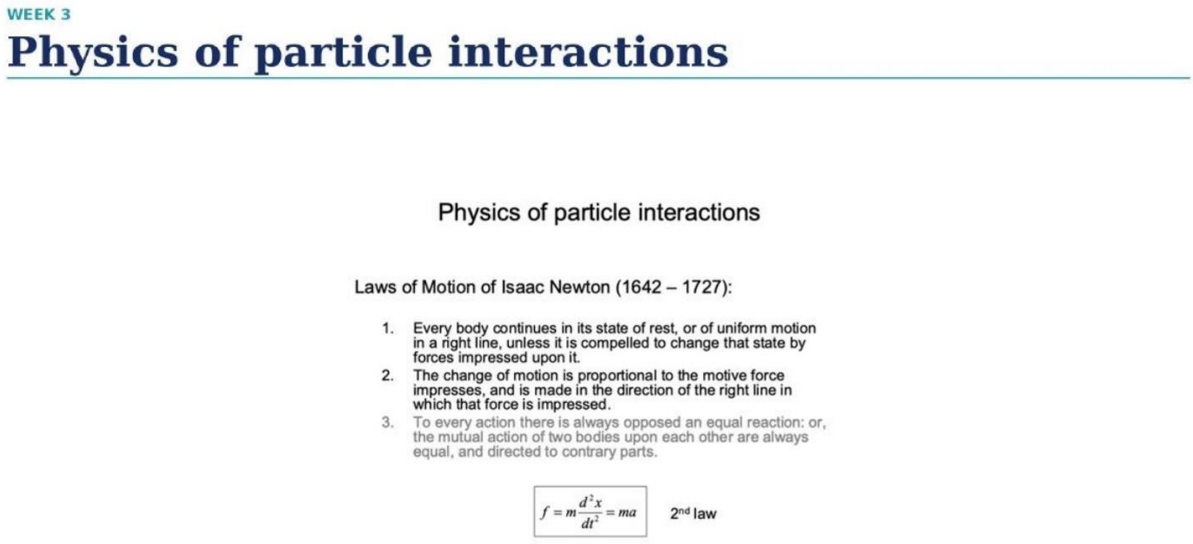
- $-r_i(t_0 - \Delta t)$ **(Red):** The previous position at time $t_0 - \Delta t$.

- $a_i(t_0)\Delta t^2$ **(Red):** The acceleration at the current time t_0.

3. Why This Is Important for Computational Modeling

- **No Velocity Needed:** Velocity doesn't need to be calculated to find the next position; only current position, previous position, and current acceleration are required.

- **Higher Accuracy:** Odd-power truncation error terms (Δt^3) drop out during addition, making the Verlet algorithm significantly more stable and accurate than simple forward Euler integration.



Slide Breakdown: Physics of Particle Interactions

This slide answers where acceleration ($a_i(t)$) comes from in the Verlet algorithm: classical Newtonian physics.

1. Newton's Laws of Motion

The slide lists Isaac Newton's three fundamental laws governing particle motion:

1. **1st Law (Inertia):** A body stays at rest or moves at a constant velocity unless acted upon by a force.

2. **2nd Law (Force & Acceleration):** The change of motion (acceleration) is proportional to the net force applied in that direction ($f = m\frac{d^2x}{dt^2} = ma$).

3. **3rd Law (Action-Reaction):** For every force exerted by one particle on another, an equal and opposite force is exerted back.

2. Connection to Molecular Dynamics & Verlet Algorithm

- **Calculating Acceleration:** In the Verlet equation ($r(t + \Delta t) = 2r(t) - r(t - \Delta t) + a(t)\Delta t^2$), acceleration $a(t)$ is required to calculate the next step.

- **The Physics Bridge:** Using Newton's 2nd Law, the computer computes acceleration directly from net force:

$$a_i(t) = \frac{f_i(t)}{m_i}$$

Once interatomic forces $f_i(t)$ are determined, acceleration $a_i(t)$ is computed, enabling the Verlet formula to advance particle positions forward in time.

WEEK 3

Verlet central difference method

Verlet central difference method

$$r_i(t_0 + \Delta t) = \underbrace{2r_i(t_0)}_{\text{Positions at } t_0} - \underbrace{r_i(t_0 - \Delta t)}_{\text{Positions at } t_0-\Delta t} + \underbrace{a_i(t_0)}_{\text{Accelerations at } t_0}\Delta t^2 + \ldots$$

How to obtain accelerations? $f_i = ma_i$, $a_i = f_i / m$

Need forces on atoms!

WEEK 3

How are forces calculated?

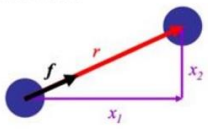
How are forces calculated?

Force magnitude: Derivative of potential energy with respect to atomic distance

$$f = -\frac{\mathrm{d}U(r)}{\mathrm{d}r}$$

To obtain force vector f_i, take projections into the three axial directions

$$f_i = f\frac{x_i}{r}$$



Often: Assume pair-wise interaction between atoms

Slide Breakdown: How Are Forces Calculated?

This slide completes the final missing link of the Molecular Dynamics loop by showing how to calculate inter-atomic forces directly from position data.

1. Force Magnitude from Potential Energy (U)

- **The Physics Rule:** Force is the negative gradient (spatial derivative) of the inter-atomic potential energy $U(r)$ with respect to distance r:

$$f = -\frac{dU(r)}{dr}$$

- **Physical Intuition:** Think of potential energy like rolling a ball down a hill. The ball experiences a force pushing it downward where the hill is steepest (the derivative $\frac{dU}{dr}$). The negative sign simply means particles are naturally pushed toward lower energy states (e.g., opposite charges attract, atoms pushed too close together repel).

2. Converting Force Magnitude to 3D Vector Components (f_i)

- Since simulations run in 3D space, the scalar force magnitude f must be projected into spatial directions (x, y, z):

$$f_i = f\frac{x_i}{r}$$

- **The Projection:** $\frac{x_i}{r}$ acts as a unit direction vector, breaking the total force magnitude into individual axial components (f_x, f_y, f_z) along the distance vector connecting the two particles.

3. Pair-Wise Interaction Assumption

- **Highlighted Note:** "Often: Assume pair-wise interaction between atoms".
- **Simplifying the Computation:** In large systems of thousands of atoms, calculating complex multi-body interactions is computationally expensive. Models assume that total force on atom A is simply the sum of individual pair-wise forces exerted on it by every neighboring atom B one-by-one.

The Complete Molecular Dynamics Computational Loop

Now all the pieces connect into a single closed loop executed millions of times per simulation:

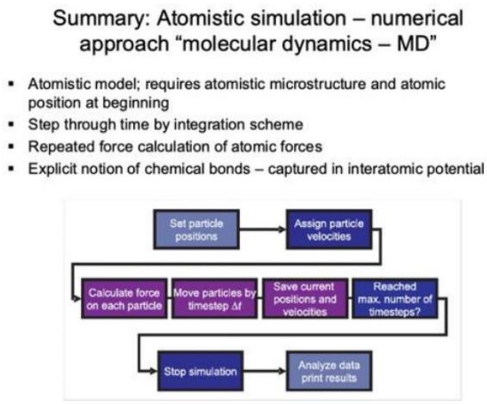
1. **Positions** $r(t)$ → Calculate inter-atomic distances r.
2. **Distances** r → Compute forces using energy derivative $f = -\frac{dU}{dr}$.
3. **Forces** $f(t)$ → Compute accelerations using Newton's 2nd law $a(t) = \frac{f(t)}{m}$.

4. **Accelerations** $a(t)$ → Plug into the Verlet Formula $r(t+\Delta t) = 2r(t) - r(t-\Delta t) + a(t)\Delta t^2$ to get new positions!

5. **Trajectories** $r(t)$ → Average over time to compute the diffusion coefficient $D = \frac{\Delta x^2}{\Delta t}$.

WEEK 3

Summary: atomistic simulation



These two slides summarize the complete computational pipeline for an **Atomistic Simulation** using **Molecular Dynamics (MD)**.

Core Principles of MD

- **Initial Configuration Required:** An atomistic model demands an initial setup of the microstructure and the exact starting position of every atom.

- **Time Stepping:** The system advances step-by-step through physical time using a numerical integration scheme (such as the Verlet algorithm).

- **Repeated Force Evaluation:** At every single step, inter-atomic forces are recalculated based on current positions.

- **Interatomic Potentials:** Chemical bonding and atomic interactions are implicitly captured via interatomic potential functions $U(r)$.

Step-by-Step Flowchart Breakdown

1. **Set Particle Positions:** Define initial coordinates $r_i(t_0)$ for all N atoms in the system grid/box.

2. **Assign Particle Velocities:** Initialize starting velocities $v_i(t_0)$ (often drawn randomly from a Maxwell-Boltzmann temperature distribution).

3. **Calculate Force on Each Particle:** Compute net pair-wise forces $f_i = -\frac{dU(r)}{dr}$ acting on each atom due to surrounding neighbors.

4. **Move Particles by Timestep Δt:** Calculate accelerations $a_i = \frac{f_i}{m_i}$ and update atomic positions to $r_i(t + \Delta t)$ using the integration algorithm.

5. **Save Current Positions and Velocities:** Record trajectory data to memory/disk.

6. **Check Loop Condition ("Reached max number of timesteps?"):**

    - **No:** Loop back to Step 3 (recompute forces for new positions).

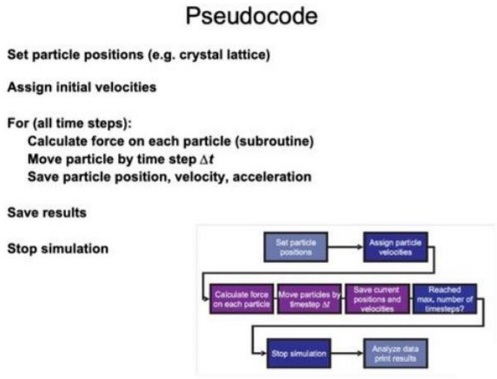
    - **Yes:** Proceed to termination.

7. **Stop Simulation:** Terminate the time-stepping loop.

8. **Analyze Data & Print Results:** Extract physical properties from saved trajectories—such as mean squared displacement to calculate $D = \frac{\Delta x^2}{\Delta t}$.

WEEK 3

Pseudocode



These two slides summarize the complete computational pipeline for an **Atomistic Simulation** using **Molecular Dynamics (MD)**.

Core Principles of MD

- **Initial Configuration Required:** An atomistic model demands an initial setup of the microstructure and the exact starting position of every atom.
- **Time Stepping:** The system advances step-by-step through physical time using a numerical integration scheme (such as the Verlet algorithm).
- **Repeated Force Evaluation:** At every single step, inter-atomic forces are recalculated based on current positions.
- **Interatomic Potentials:** Chemical bonding and atomic interactions are implicitly captured via interatomic potential functions $U(r)$.

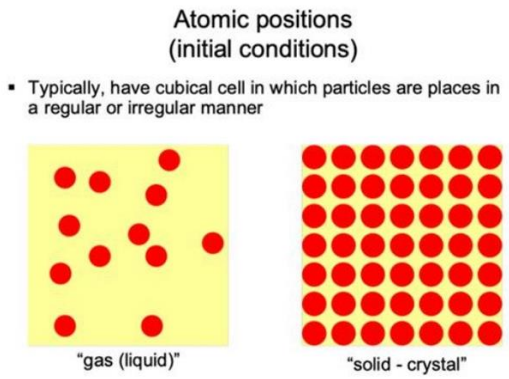
Step-by-Step Flowchart Breakdown

1. **Set Particle Positions:** Define initial coordinates $r_i(t_0)$ for all N atoms in the system grid/box.

2. **Assign Particle Velocities:** Initialize starting velocities $v_i(t_0)$ (often drawn randomly from a Maxwell-Boltzmann temperature distribution).
3. **Calculate Force on Each Particle:** Compute net pair-wise forces $f_i = -\frac{dU(r)}{dr}$ acting on each atom due to surrounding neighbors.
4. **Move Particles by Timestep Δt:** Calculate accelerations $a_i = \frac{f_i}{m_i}$ and update atomic positions to $r_i(t + \Delta t)$ using the integration algorithm.
5. **Save Current Positions and Velocities:** Record trajectory data to memory/disk.
6. **Check Loop Condition ("Reached max number of timesteps?"):**
    - **No:** Loop back to Step 3 (recompute forces for new positions).
    - **Yes:** Proceed to termination.
7. **Stop Simulation:** Terminate the time-stepping loop.
8. **Analyze Data & Print Results:** Extract physical properties from saved trajectories—such as mean squared displacement to calculate $D = \frac{\Delta x^2}{\Delta t}$.

WEEK 3

Atomic positions



Slide Breakdown: Atomic Positions (Initial Conditions)

This slide details **Step 1** of the Molecular Dynamics flowchart: establishing the initial coordinates $r_i(t_0)$ for all particles in the system before time begins.

1. The Simulation Cell Setup

- **Cubical Cell:** Simulations are set up inside a defined 3D box (simulation cell or domain) containing all the target particles.
- **Placement Modes:** Particles are placed inside this cell in either a **regular** (structured lattice) or **irregular** (randomized) arrangement, depending on the phase of matter being studied.

2. Phase-Specific Initial Configurations

- **Gas / Liquid (Irregular Manner):**
    - Particles are placed randomly or with loose spacing throughout the box.
    - Represents fluid environments where atoms move freely without fixed crystal lattices (e.g., drug molecules dissolved in water or blood plasma).

- **Solid - Crystal (Regular Manner):**

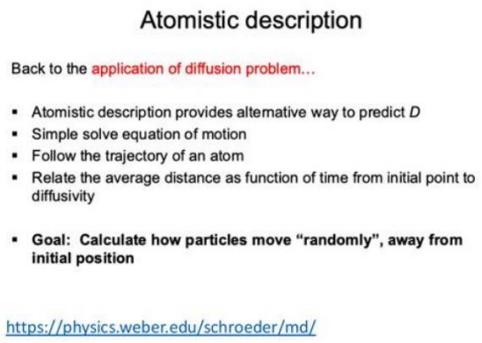
- Particles are placed on a structured geometric grid (such as Face-Centered Cubic or Body-Centered Cubic lattices).
- Represents structured solid states where atoms vibrate around fixed lattice positions (e.g., metals, ceramics, or rigid biomaterial scaffolds).

3. Why Initial Positions Matter

- To start the Verlet algorithm, the computer needs an initial spatial distance r between every pair of atoms.
- These starting distances determine the initial forces ($f = -\frac{dU}{dr}$) and initial accelerations ($a = \frac{f}{m}$) that kick off the first integration step.

WEEK 3

Atomistic description



This slide brings the focus back to the primary application: predicting the diffusion coefficient D directly from an atomistic model.

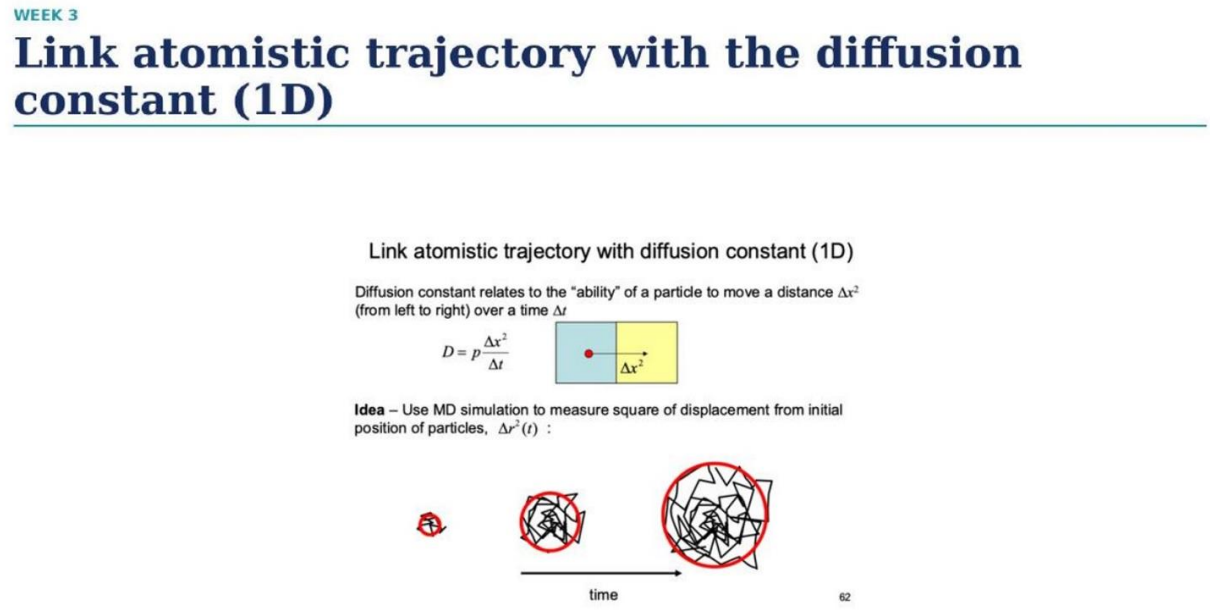
Key Points

- **Alternative Route to D:** The atomistic description offers a direct computational method to predict D without needing prior laboratory curve-fitting.
- **Equations of Motion:** It achieves this by numerically solving Newton's equations of motion ($f = ma$) using algorithms like Verlet to update particle positions step-by-step.

- **Tracking Trajectories:** The simulation follows the exact trajectory $r_i(t)$ of an individual atom or molecule as it moves through space over time.
- **Relating Distance to Diffusivity:** It computes D by relating the average distance squared particles travel from their initial starting points over time.
- **Core Goal:** Quantify how particles move "randomly" away from their initial positions to extract microscopic transport behavior.

Summary of the Atomistic Workflow

1. **Initialize:** Set initial atomic positions $r_i(t_0)$ (crystal lattice or gas/liquid) and velocities $v_i(t_0)$.
2. **Compute Forces & Accelerations:** Calculate inter-atomic forces $f = -\frac{dU}{dr}$ to find accelerations $a = \frac{f}{m}$.
3. **Integrate Trajectories:** Step forward in time using the Verlet algorithm: $r(t + \Delta t) = 2r(t) - r(t - \Delta t) + a(t)\Delta t^2$.
4. **Extract Diffusivity:** Calculate how far particles drift from their origin over time to yield $D = \frac{\Delta x^2}{\Delta t}$.



Slide Breakdown: Link Atomistic Trajectory with the Diffusion Constant (1D)

This slide connects the micro-scale particle trajectories generated by Molecular Dynamics (MD) directly to the macro-scale diffusion constant D.

1. Microscopic Definition of D

- **Physical Intuition:** The diffusion constant measures the "ability" of a particle to displace a squared distance Δx^2 over a time interval Δt.

- **1D Formula:**

$$D = p\frac{\Delta x^2}{\Delta t}$$

(where p represents jump probability, Δx is the step length, and Δt is the time step).

2. The Core Idea: Measuring Displacement via MD Trajectories

- **Tracking Random Walks:** As time progresses, a particle moves randomly (represented by the black squiggly line trajectory) away from its starting location.
- **Expanding Boundary:** The red circles show how the spatial region explored by the particle grows over time.
- **Square of Displacement ($\Delta r^2(t)$):** Instead of measuring single discrete steps, MD records the cumulative **squared distance** between the particle's current position and its origin at time t:

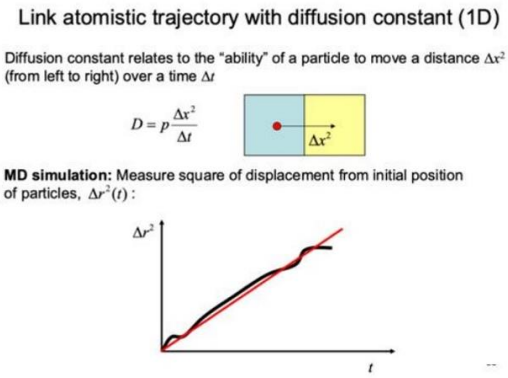
$$\Delta r^2(t) = |r(t) - r(0)|^2$$

3. How This Completes the Simulation Bridge

1. **Verlet Integration** tracks $r(t)$ step-by-step using inter-atomic forces.
2. **MD Outputs** the spatial trajectory of all particles over time.
3. **Statistical Averaging** calculates how fast $\Delta r^2(t)$ grows on average over time to yield the exact value of D!

WEEK 3

Link atomistic trajectory with the diffusion constant (1D)



Slide Breakdown: Link Atomistic Trajectories with Diffusion Constant (1D)

This slide illustrates how the trajectory output from a Molecular Dynamics (MD) simulation is converted into a numerical value for the diffusion constant D.

1. Core Mathematical Principle

- **Diffusivity Definition:** D represents a particle's physical ability to cover a squared distance Δx^2 within a given time frame Δt:

$$D = p\frac{\Delta x^2}{\Delta t}$$

2. Plotting Mean Squared Displacement ($\Delta r^2(t)$ vs. Time t)

- **Data Extraction:** The MD simulation tracks particle coordinates over thousands of timesteps and computes the **squared displacement from the initial position**, $\Delta r^2(t)$.
- **The Graph:**
    - **Black Squiggly Line:** The raw, instantaneous squared displacement measured from individual atom trajectories over time t.

- **Red Straight Line:** The linear best-fit line calculated across the simulation duration.

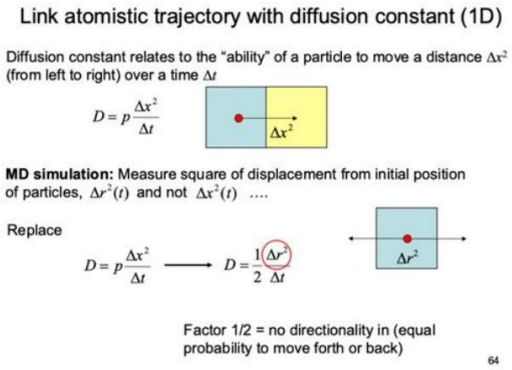
3. Extracting D from the Slope

- **Linearity of Brownian Motion:** Over time, the mean squared displacement grows linearly with time (t).
- **Calculating Diffusivity:** The slope of that linear red trendline directly equals the rate of spatial spreading over time ($\frac{\Delta r^2}{\Delta t}$).
- **The Result:** Dividing the slope by the dimension-scaling factor gives the microscopic diffusion coefficient D, completing the bottom-up MD pipeline!

We use the linear fit (red line) instead of the raw data (black line) because diffusion is a long-term statistical average, whereas the instant fluctuations in the raw data are just short-term atomic collisions and thermal noise.

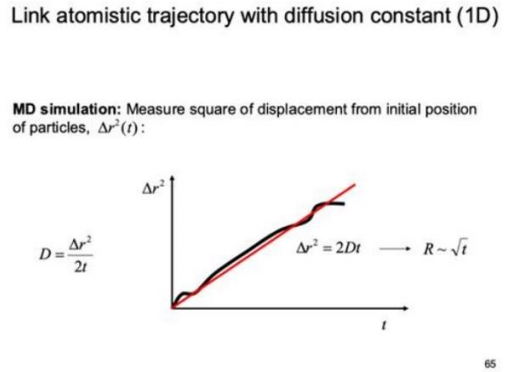
WEEK 3

Link atomistic trajectory with the diffusion constant (1D)



WEEK 3

Link atomistic trajectory with the diffusion constant (1D)



Slide Breakdown: Link Atomistic Trajectory with Diffusion Constant (1D)

These two slides show the exact mathematical transition from tracking absolute particle displacements to computing the 1D diffusion coefficient D in a Molecular Dynamics (MD) simulation.

1. Slide 64 Breakdown: Transitioning to Unbiased Displacement (Δr^2)

- **The Distinction:** Instead of tracking directional step sizes (Δx^2), MD measures total radial displacement squared from the origin, $\Delta r^2(t)$.

- **Replacing Terms:**

$$D = p\frac{\Delta x^2}{\Delta t} \longrightarrow D = \frac{1}{2}\frac{\Delta r^2}{\Delta t}$$

- **The Physical Meaning of $\frac{1}{2}$:** In 1D random movement, a particle has equal probability of stepping left or right (no directionality). The factor of $\frac{1}{2}$ accounts for this isotropic 1D symmetry.

2. Slide 65 Breakdown: The Einstein Relation in 1D

- **Rearranging the Formula:** Multiplying both sides of $D = \frac{\Delta r^2}{2t}$ by $2t$ yields the classical 1D Einstein relation:

$$\Delta r^2 = 2Dt$$

- **Extracting D from the MSD Plot:**

  - The slope of the linear fit line (red) on the Δr^2 vs t plot equals $2D$.
  - Therefore, diffusivity is calculated as:

$$D = \frac{\text{Slope}}{2}$$

- **Spatial Spreading Scaling ($R \sim \sqrt{t}$):**

  - Taking the square root of both sides gives the root-mean-square displacement: $R = \sqrt{\Delta r^2} = \sqrt{2Dt} \propto \sqrt{t}$.

- **Biological/Physical takeaway:** A diffusing particle's net distance grows with the **square root of time** ($\sqrt{t}$), which is why simple diffusion is extremely fast over sub-micron cellular distances but inefficient over macro-scale distances.

Yes, Δx and Δr differ in both **definition** and **spatial meaning** within the derivation:

1. Δx **is a single discrete jump step along one direction**. In the microscopic random-walk model, Δx represents the fixed step length a particle takes (left or right) during a single jump interval Δt.

2. $\Delta r(t)$ **is the cumulative displacement vector from the origin**. In a Molecular Dynamics (MD) simulation, $\Delta r(t) = |r(t) - r(0)|$ measures the total net distance the particle has traveled away from its starting position after time t.

Why the Formula Changes

- **Local Jump Model:**

$$D = p\frac{\Delta x^2}{\Delta t}$$

Here, p is the probability of a particle jumping specifically toward the right ($p = \frac{1}{2}$ for unbiased 1D movement).

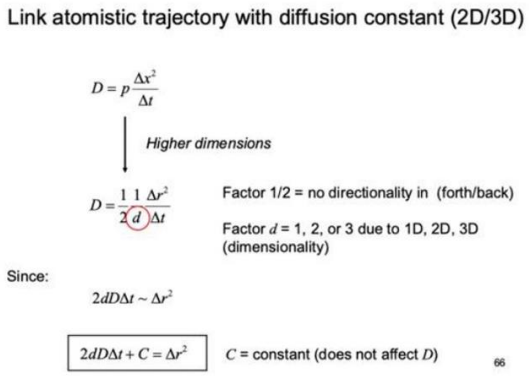
- **MD Measured Model:**

$$D = \frac{1}{2}\frac{\Delta r^2}{\Delta t}$$

Instead of tracking individual microscopic jumps Δx, MD tracks the cumulative squared distance $\Delta r^2(t)$. Replacing p with $\frac{1}{2}$ reflects that in 1D, a particle has equal probability of moving forward or backward, removing any directional bias from the overall spatial spreading.

WEEK 3

Link atomistic trajectory with the diffusion constant (2D/3D)



Slide Breakdown: Link Atomistic Trajectory with Diffusion Constant (2D/3D)

This slide generalizes the 1D Einstein relation to higher spatial dimensions (2D and 3D).

1. Generalizing the Formula for Higher Dimensions (d)

Starting from the basic 1D jump relationship $D = p\frac{\Delta x^2}{\Delta t}$, expanding to d spatial dimensions yields:

$$D = \frac{1}{2d}\frac{\Delta r^2}{\Delta t}$$

- **Factor** $1/2$**:** Accounts for no directionality bias in movement (equal probability to move forward or backward along any single axis).

- **Dimensionality Factor** d**:** Represents the number of spatial degrees of freedom ($d = 1,2,$ or 3):

- **1D ($d = 1$):** $D = \frac{\Delta r^2}{2\Delta t} \Longrightarrow \Delta r^2 = 2Dt$

- **2D ($d = 2$):** $D = \frac{\Delta r^2}{4\Delta t} \Longrightarrow \Delta r^2 = 4Dt$ (e.g., cell membrane diffusion)

- **3D ($d = 3$):** $D = \frac{\Delta r^2}{6\Delta t} \Longrightarrow \Delta r^2 = 6Dt$ (e.g., bulk liquid or biological tissue diffusion)

2. The Linear Fit Equation with Constant C

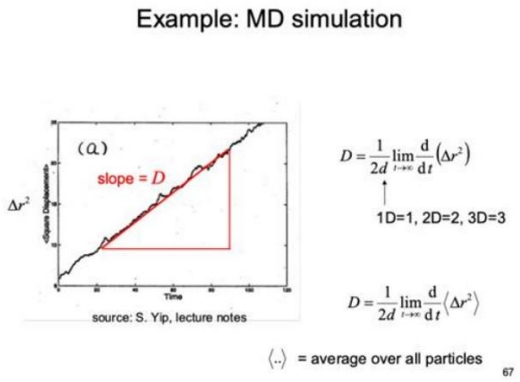
Rearranging the formula gives the full equation of a line used to extract D from simulation data:

$$\Delta r^2 = 2dD\Delta t + C$$

- **Slope =** $2dD$**:** The slope of the long-time linear trendline of Δr^2 vs t is divided by $2d$ to calculate D.
- **Integration Constant** C**:** Represents the y-intercept. It captures the short-time non-diffusive (ballistic) motion and local atomic vibration offsets, but does **not** affect the value of D, since D is strictly determined by the slope.

WEEK 3

Example: MD simulation



Slide Breakdown: Example: MD Simulation

This slide illustrates the final step of extracting the diffusion constant D from actual Molecular Dynamics (MD) trajectory data using the generalized Einstein-Smoluchowski relation.

1. Long-Time Derivative & Limit Definition

- **Mathematical Expression:**

$$D = \frac{1}{2d}\lim_{t\to\infty}\frac{d}{dt}\langle\Delta r^2\rangle$$

- **The Long-Time Limit ($\lim_{t\to\infty}$):** Diffusivity must be evaluated over long simulation times to ensure the particle has transitioned out of the initial ballistic (short-time collision) regime into pure diffusive motion.

- **The Time Derivative ($\frac{d}{dt}$):** Formally takes the slope of the mean squared displacement relative to time.

2. Statistical Ensemble Averaging ($\langle\dots\rangle$)

- **Bracket Notation ($\langle\Delta r^2\rangle$):** Denotes the average over all particles in the simulation system.

- **Why Average Across Particles?** Tracking a single particle yields high statistical noise. Averaging the squared displacement across hundreds or thousands of identical particles smooths out random fluctuations, giving a clean linear trendline.

3. Dimensionality Scaling (d)

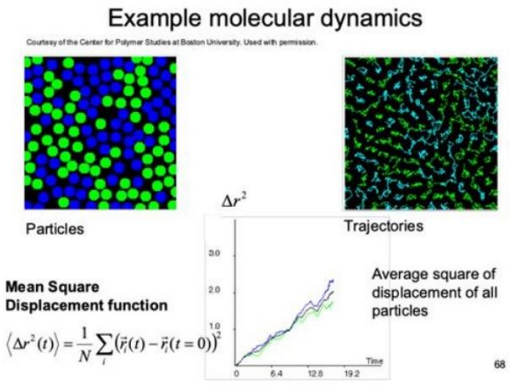
- **Spatial Dimension Factor (d):**
  - **1D ($d = 1$):** $D = \frac{1}{2}\frac{d}{dt}\langle \Delta r^2 \rangle$
  - **2D ($d = 2$):** $D = \frac{1}{4}\frac{d}{dt}\langle \Delta r^2 \rangle$
  - **3D ($d = 3$):** $D = \frac{1}{6}\frac{d}{dt}\langle \Delta r^2 \rangle$

4. Practical Implementation from the Graph

- **Plotting:** Mean Squared Displacement ($\langle \Delta r^2 \rangle$) is plotted on the y-axis against physical Time on the x-axis.
- **Calculating Slope:** The red triangle shows the linear fit over the linear region.
- **Obtaining D:** The measured slope ($\frac{\Delta r^2}{\Delta t}$) is divided by $2d$ to obtain the macroscopic diffusion constant D used in continuum engineering transport models!

WEEK 3

Example molecular dynamics



Slide 68: Example Molecular Dynamics

- **Particle Systems & Trajectories:** Visualizes a system of particles (green spheres) moving through space, tracking their individual random-walk paths over time to form complex trajectory maps.

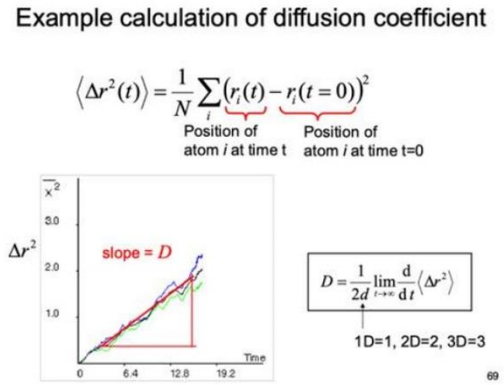
- **Ensemble Mean Square Displacement (MSD):** Formally defines the MSD equation by taking the arithmetic mean of squared distances across all N particles in the system:

$$\langle \Delta r^2(t) \rangle = \frac{1}{N} \sum_i \big(r_i(t) - r_i(t=0)\big)^2$$

- **Plotting MSD vs. Time:** Shows multiple colored lines representing MSD data tracked across simulation runs, illustrating how net spatial displacement grows linearly over time.

WEEK 3

Example calculation of the diffusion coefficient



Slide 69: Example Calculation of the Diffusion Coefficient

- **Position Tracking:** Re-emphasizes the displacement of each atom i from its starting position $r_i(t = 0)$ to its position at time t, $r_i(t)$.

- **Slope Extraction:** Fits a linear red trendline to the long-time MSD data plot to determine its slope ($\frac{d}{dt}\langle \Delta r^2 \rangle$).

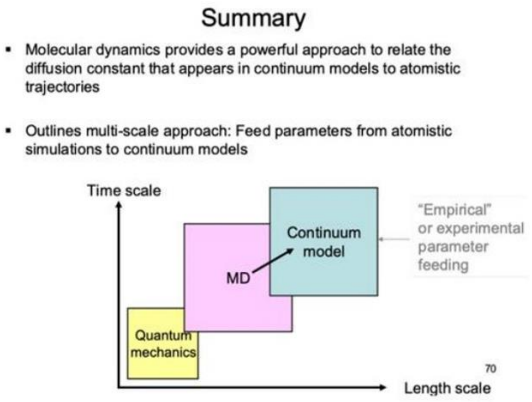
- **Einstein Relation Calculation:** Applies the multi-dimensional Einstein relation to solve for D:

$$D = \frac{1}{2d} \lim_{t\to\infty} \frac{d}{dt} \langle \Delta r^2 \rangle$$

(where $d = 1,2,$ or 3 depending on spatial dimensions).

WEEK 3

Summary



Slide 70: Summary (Multiscale Modeling Framework)

- **Bottom-Up Parameterization:** Molecular Dynamics bridges atomistic trajectory simulations to macro-scale continuum engineering transport models by directly calculating bulk properties like the diffusion constant D.

- **Multiscale Hierarchy:** Maps out physical modeling domains across spatial and temporal scales:

    - **Quantum Mechanics:** Electronic scale (femtoseconds / angstroms).

    - **Molecular Dynamics (MD):** Atomistic scale (nanoseconds / nanometers).

    - **Continuum Models:** Macro engineering scale (seconds / millimeters to meters).

- **Eliminating Empirical Fitting:** Instead of relying purely on experimental curve-fitting, MD directly feeds physical transport parameters into continuum models from first principles.